# 11. Moon Births

*One mechanism for the births of moons, tested planet by planet across the Solar System*

Zohar Duran · ORCID 0009-0005-3194-9387 · 9 November 2026

**Abstract**

The Moon was built from Earth's material, on the tip of a pear that the young, fast-spinning Earth raised out of itself, and it was released when its centre crossed the corotation line – the height at which a body rotating together with Earth moves at orbital velocity [Papers 2–3]. On Mars, a sharpening bulge raised volcanoes, and Phobos and Deimos were ejected from them [Paper 10]. This paper defines what the two cases have in common and formulates a uniform mechanism for the birth of moons: the conditions required for the process to take place on a planet, and the traces it leaves on the parent and on the moon.

**The mechanism rests on two laws and two physical principles:**

- **The birth law:** a moon is born when the parent's corotation line stands at a position that the mode of birth allows to be reached. The required distance is about 2 radii for a pear and a bulge, close to the surface for a volcano without a bulge, and about 1 for a split. The position of the line in radius units depends solely on the ratio between spin rate and density, and is therefore applied uniformly to all bodies. The birth day follows from this and depends only on the density of the parent.
- **The birth window:** the process is possible only at a defined stage in the life of the parent. The window is set by the stage of differentiation, not by a calendar measure of years.
- **The pump principle:** the differentiation process pushes water and light materials from the interior of the body toward the exit opening. The window stays open as long as the pump operates and has an outlet.
- **The cap (lid) principle:** the distance from the Sun set the amount of hydrogen and helium gathered before the disc dispersed. A thick gas envelope closes over the water and leaves the corotation line in place. A thin envelope allows the water to escape through the line, carry angular momentum and push the line outward.

**Moons are born in three modes, set by the position of the parent in the solar system:**

**1. The inner edge:** near the Sun, the parent dries quickly and holds no water for ejection. Earth's Moon is the only one built on the tip of a pear, and it is therefore dry and large relative to its parent (1.2%).

**2. The central zone:** from Mars to Neptune, the bodies are large and wet. The moons were ejected with large amounts of water, and they are very small relative to the parent (0.01%–0.02% in the gas and ice giants).

**3. The outer edge:** Pluto and Charon are one body that split. The united body spun faster than the elongation threshold, was stretched into a peanut shape, and its differentiation accelerated it beyond the separation threshold until it was cut through at its neck.

The differentiation process brings the corotation line closer to the surface: the parent contracts in its interior, spins up, and swells in its outer part. The composition of the material ejected in each birth is set by the stage of differentiation:

- **At the start of the process:** material is ejected that directly represents the composition of the parent.
- **At the peak of the pump's activity:** water-laden material is ejected that consolidates into a spongy body.
- **At the end of the process:** a dense, dry residue is ejected.

The mechanism is applied as a uniform protocol of 13 checks to all the planets and to five dwarf planets. Mercury and Venus lack several matching conditions, and are therefore moonless. Jupiter and Saturn rotate today within the birth range, and their thick gas envelope closes over the water. On Saturn, water that escapes is trapped in the rings and returns as ring rain, the ring being a continuous flow. Uranus and Neptune left the birth range because of water escaping through the line; their inner moons are located to this day within the birth range and are falling back.

In five bodies a fixed pattern recurs in which small moons are located on both sides of the corotation line. Moon density decreases as the distance from the Sun increases: the density of Earth's Moon is 3,344 kg/m³, the Galilean moons about 2,600 kg/m³, and Saturn's moons about 1,350 kg/m³. Uranus, which is almost twice as dense as Saturn, gives birth to denser moons. The chain of links is defined as: distance from the Sun $\rightarrow$ cap $\rightarrow$ parent density $\rightarrow$ moon density. On Pluto, the small moons were ejected from

the equator before the split and moved away ahead of the line, which never caught up with them. The common-denominator matrix (Part L) shows no bodies that contradict the mechanism; it points to locations that require further research, and Part M gathers them as open questions.

**What this paper proposes**

- **Claim:** the birth of moons is a uniform process throughout the solar system. Fast spin of a soft, wet parent raises material out of it (as a pear, a bulge or the whole body) up to the corotation line, and the material that crosses the line becomes a moon. The differences between the systems arise from variation in density, spin, amount of water, stage of differentiation and distance from the Sun, not from separate mechanisms. These factors converge into a single law – the moon law (A.10).
- Formulas and numerical values:
  - Position of the corotation line in radius units ($x$):

$$x^3 = \frac{4\pi G\rho}{3\Omega^2}$$

  - The ruler of birth modes ($x$): pear: 2.29, bulge: 1.74–2.06 (based on measurements on Earth and Mars), adjacent volcano: about 1.1–1.5, split: about 1.
  - Approach of the line through inner contraction and outer swelling:

$$x \propto (k^2 R)^{1/3}$$

  - The birth day ($P_{\mathrm{birth}}$) for $x \approx 2$:

$$P_{\mathrm{birth}} = \sqrt{\frac{3\pi x^3}{G\rho}}$$

  - **Three modes of birth:** built (Earth's Moon, 1.2% of the parent's mass), ejected (0.01%–0.02% in the giants), split (Charon, 11% of the pair's mass).
- Body-specific requirements (Part N):
  - **Mars:** loss of 85%–88% of young Mars's angular momentum through the ring and the escaped water.
  - **Uranus and Neptune:** escape of water from the line amounting to about 2%–3% of the planet's mass.
  - **Saturn:** a rate of water leakage into the ring equal to the rate of ring rain.

- **Pluto:** pushing of the small moons outward during the peanut stage.
- **Jupiter and Uranus:** pushing of the distant moons outward.
- **The giants (Jupiter, Saturn, Uranus, Neptune):** birth from the inner body and not from the gas envelope.

• **Tests:** application of 13 fixed checks to each planet (Part B), whose results are gathered in a single matrix (Part L).

`The setting`

### Terms

| Term | Scientific definition |
|---|---|
| Corotation line (synchronous radius) | The height at which a body orbiting the parent completes an orbit in the same time as the parent's rotation about its axis. A body below the line orbits faster than the rotation, and a body above it orbits more slowly. |
| Pear | An elongation of the whole planet, in which the inner material moves and the outer layers ride on it [Paper 3]. |
| Bulge | A local elongation on the equator, in which the outer material moves outward [Paper 10]. |
| Hill radius | The region in which the gravity of the planet dominates over the gravity of the Sun. |
| Roche limit | The distance below which the tidal forces of the parent break up a body and prevent it from consolidating. |

### Data and units

• **Data sources:** masses, radii, densities, rotation periods and orbits are taken from the NASA fact sheets (Williams 2024) and from the JPL satellite parameters (JPL Solar System Dynamics 2024).

• **Regular moon:** defined as a moon orbiting on a close orbit, in the direction of the parent's rotation and in the plane of its equator.

• **Captured moon:** defined by a points rule (L.3); not included in the protocol checks.

• **Units:** distances are given in km or in units of the parent's radius ($R$) from the centre. Density is given in kilograms per cubic metre ($\mathrm{kg/m^3}$). "Day" is defined as the sidereal

> rotation period (Sidereal rotation period).

## PART A The uniform mechanism

This part formulates the scientific mechanism against which all the planets are examined. The mechanism is built from five kinds of conditions: material, spin and geometry, the physical mechanism, survival, and geophysical signatures. These conditions are summarised by a single structural parameter and by the stages of differentiation that set the composition of the ejected material, and they converge into one physical law – the moon law. All the conditions are derived from two earlier test cases: Earth and the Moon [Papers 1–9] and Mars and its moons (Phobos and Deimos) [Paper 10].

### A.1 Introduction: from two cases to a rule

In both test cases an identical mechanical process took place in a uniform chronological sequence: a young planet, of low viscosity and high spin rate, raised inner material toward the corotation line. The material that crossed the line consolidated into a moon, while the parent continued its differentiation and slowed down, until the possibility of birth was blocked.

**The difference between the cases comes down to the mechanical character of the construction:**

- **Earth:** the Moon was built on the tip of a pear (Pear-shaped equilibrium figure). Material accumulated at the tip of the structure over time, the Moon swelled at its site of formation, and it detached when its centre of mass crossed the corotation line [Paper 3]. The mass of the original Moon is 1.2% of Earth's mass.
- **Mars:** the bulge raised volcanic structures, and from them the moons Phobos and Deimos were ejected through a point-injection mechanism [Paper 10]. For this reason, the moons of Mars are tiny relative to the parent.

The formulated mechanism is tested as a general law: a planet that satisfies the required set of conditions will give birth to a moon, whereas a planet that does not satisfy them will remain moonless. A planet that gave birth to a moon will display the defined structural and dynamical signatures.

### A.2 Material conditions

• **Material sorted outward:** the material that rises in the pear structure or is ejected from the volcanic structure is light sialic and silicate material, pushed outward in the planet's sorting process. In the outer solar system, ice joins this component. The moon is composed mainly of this material, but contains remnants of heavy elements (iron and metals) at a concentration that depends on the parent's stage of differentiation at the moment of birth. Earth's Moon has an iron core of between 1% and 2% of its mass [Paper 3].

• **Partial sorting of the parent:** a parent that has completed differentiation cannot give birth to moons. In a state of full differentiation, the core and the layers are rigidly organised (heavy material inside and light outside) and the viscosity is high, which prevents the internal hydrodynamic motion needed to form a pear. Birth requires a stage in which differentiation is active but has not yet reached its end.

• **Water and low viscosity:** the presence of water lowers the viscosity of the material and the overall density of the parent. Low viscosity allows stretching into a pear structure or flow in volcanic conduits, whereas viscous, rigid material prevents this.

• **Low density:** a low-density body has a larger radius for a given mass. As a result, surface gravity is weaker, and the surface is geometrically close to the corotation line.

### A.3 Spin and geometry conditions

• **Corotation line (synchronous radius):** the exact height at which an orbiting body circles the parent at the parent's own spin rate is defined by the equation:

$$a_c = \left( \frac{GMP^2}{4\pi^2} \right)^{1/3}$$

where $M$ is the parent's mass and $P$ is its spin period (the day). A high spin rate brings the line closer to the surface. Crossing this line is essential, since below it the material is rotationally locked to the parent, and only beyond it does it acquire an independent orbital velocity.

• **The birth law:** Earth and Mars parameters: at the moment of birth, the corotation line stood at a distance of about 2 radii from the parent's centre in both calculated cases:

  • **Earth:** for a day of 5.77 hours and an expanded radius of 7,157 km, the corotation line stands at 16,400 km, which is $2.29R$ [Paper 2].

  • **Mars:** for a day of 7–9 hours and a radius expanded by a factor of 1.5 (5,085 km), the corotation line stands in the range 8,800–10,400 km, which is $1.74R - 2.05R$ [Paper 10].

In both cases the distance of the line from the surface was about one radius, a distance that a pear or bulge structure can reach.

- **Dependence of the line's position on a single parameter:** we define the position of the line in units of the parent's radius as $x = a_c/R$. Substituting the mass as a function of the mean density ($M = \frac{4}{3}\pi R^3\rho$) into the line formula cancels the radius $R$:

$$x^3 = \frac{4\pi G\rho}{3\Omega^2}, \quad \Omega = \frac{2\pi}{P}$$

The position of the line in radius units depends solely on the ratio between the axial spin rate ($\Omega$) and the mean density ($\rho$), independent of the body's absolute dimensions. This ratio ($\frac{\Omega^2}{\pi G\rho}$) also sets the degree of oblateness, the threshold for transition to an elongated (peanut) shape and the break-up threshold of a soft rotating body [Chandrasekhar 1969].

| **Body at the moment of birth** | **Density $\rho$ (kg/m³)** | **Day $P$ (hours)** | **$\Omega \times 10^{-4}$ (per second)** | $x^3$ | $x$ |
|---|---|---|---|---|---|
| Earth | 3,940 | 5.77 | 3.02 | 12.0 | 2.29 |
| Mars | 1,170 | 7–9 | 2.49–1.94 | 5.26–8.70 | 1.74–2.06 |
| Earth today (for comparison) | 5,513 | 23.9 | 0.729 | 290 | 6.62 |

- **The ruler of birth modes:** on the basis of the rotating-body equation, rigid structural thresholds were set. At $x = 1.53$ the body passes into an elongated shape (peanut structure). At $x = 1$ the equator reaches orbital velocity and the body breaks apart (split).

| **Mode of birth** | **Mechanism raising the material** | **Position of the line ($x$)** | **Physical parameters then** | **Source of value** | **Example** |
|---|---|---|---|---|---|
| Built | Pear structure | 2.29 | 5.77 hours; 3,940 kg/m³ | Calculated [Paper 2] | Earth's Moon |
| Ejected on a bulge | Bulge and volcano | 1.74–2.06 | 7–9 hours; 1,170 kg/m³ | Calculated [Paper 10] | Phobos and Deimos |
| Ejected without a bulge | Volcano or local hill | about 1.1–1.5 | — | Theoretical possibility | No calculated example |

| | | | | | |
|---|---|---|---|---|---|
| Split | Cutting through the whole body | about 0.85–1.02 | 1.9–2.5 hours; 1,840 kg/m³ | Calculated [K.1] | Pluto and Charon |

Statement of the birth law: a moon is born when the parent's corotation line stands at a position matching the mechanical mode of birth: about 2 radii ($1.75R - 2.3R$) for a pear and a bulge, close to the surface ($1.1R - 1.5R$) for a volcano without a bulge, and about 1 radius for a full split.

- **Dependence of the birth day on density alone:** solving the equation for the spin period $P$ under the assumption $a_c = xR$ shows a dependence solely on the mean density $\rho$:

$$P_{\text{birth}} = \sqrt{\frac{3\pi x^3}{G\rho}}$$

A low-density body needs a lower spin rate to reach the birth threshold, whereas a dense body needs a significantly higher spin rate.

- **Critical rate:** the lower threshold at which the equator reaches orbital velocity and causes the body to break apart is defined at $x = 1$:

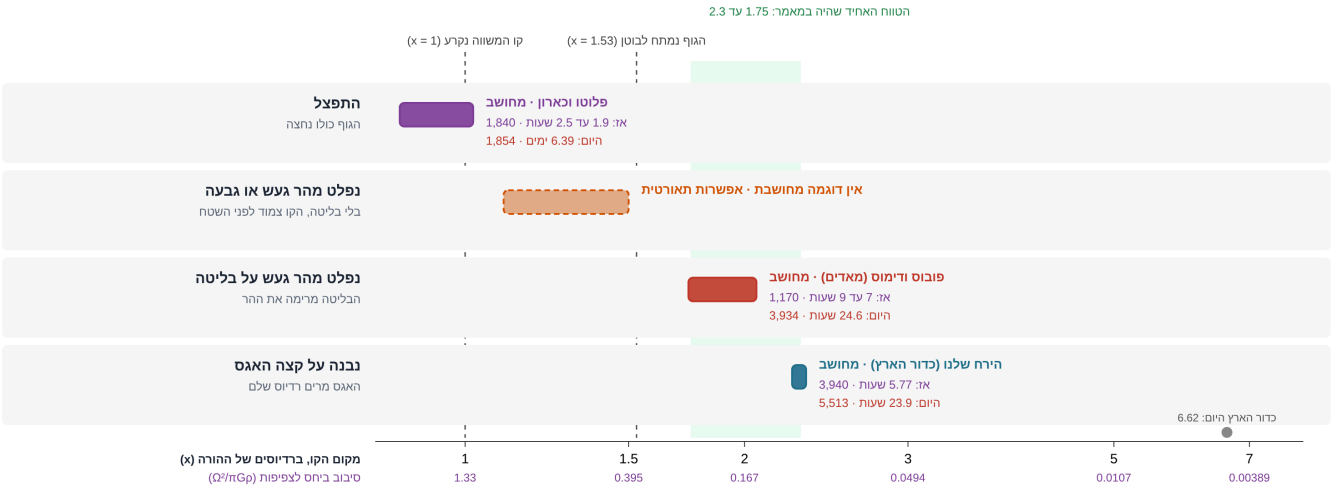
$$P_{\text{critical}} = \sqrt{\frac{3\pi}{G\rho}}$$

Birth from a pear or a bulge ($x \approx 2$) takes place at a day 2.8 times longer than the critical day, and does not require the planet to break apart, only that the line come closer. A full split takes place exactly at this threshold.

- **Flattening and axial tilt:** fast spin causes oblateness at the poles and a bulge at the equator, which reduces the local gravity component and accelerates the concentration of material at the equator. The plane of the parent's equator at the moment of birth sets the plane of the moon's orbit, and therefore regular moons move in the plane of the planet's geometric equator.
- **Angular momentum:** the total angular momentum of the system (the parent's spin and the orbital momentum of the moons) is conserved, in the absence of a mechanism for external momentum escape. This conservation allows the original spin rate of the parent to be reconstructed backward (protocol check number 3).

## סרגל אופני הלידה: איפה עומד קו הקורוטציה בכל אופן

כל שורה היא אופן לידה. הפס מראה איפה עמד קו הקורוטציה ברגע הלידה, ברדיוסים של ההורה. ליד כל פס: היממה והצפיפות אז (סגול) והיום (אדום). פס מלא: מחושב בסדרה. פס מקווקו ושקוף: אפשרות תאורטית, בלי דוגמה מחושבת.




**הנוסחה:** $x^3 = 4\pi G\rho / (3\Omega^2)$, כאשר $\Omega = 2\pi / P$. מקום הקו ברדיוסים תלוי רק בצפיפות ובמהירות הסיבוב, ולא בגודל הגוף.

| גוף | צפיפות ρ (ק"ג למ"ק) | יממה P (שעות) | Ω (×10⁻⁴ לשנייה) | x³ | x |
|---|---|---|---|---|---|
| הירח שלנו | 3,940 | 5.77 | 3.02 | 12.0 | **2.29** |
| פובוס ודימוס | 1,170 | 7 עד 9 | 2.49 עד 1.94 | 5.26 עד 8.70 | **1.74 עד 2.06** |
| פלוטו וכארון | 1,840 | 2.5 עד 1.9 | 6.98 עד 9.19 | 1.05 עד 0.61 | **1.02 עד 0.85** |
| כדור הארץ היום | 5,513 | 23.9 | 0.729 | 290 | **6.62** |

ככל שהקו קרוב יותר לפני השטח, נדרשת פחות הרמה: האגס מרים רדיוס שלם, הבליטה פחות, הר געש כמעט כלום, ובפיצול אין הרמה כלל. הקווים האפורים המקוקווים הם ספי הצורה של גוף רך מסתובב: מתחת ל-1.53 הגוף אינו יכול להישאר עגול ונמתח לבוטן; ב-1 קו המשווה נקרע. בפלוטו וכארון x יורד מתחת ל-1 כשהגוף המאוחד מתמיין: אותו תנע זוויתי, מסה מרוכזת יותר, סיבוב מהיר יותר.

**Figure A.1.** The ruler of birth modes. Each row is a mode of birth, and the bar shows where the corotation line stood at the moment of birth, in units of the parent's radius. Next to each bar the day and the density then and today are given. Solid bar: calculated; dashed bar: estimate. The lower axis gives the position of the line and, beneath it, the ratio between spin and density; the grey lines are the shape thresholds. The green band is the pear and bulge range. At the bottom: the formula and the full numbers. *How to read:* the closer the bar is to 1, the closer the line was to the surface and the less lifting was needed.

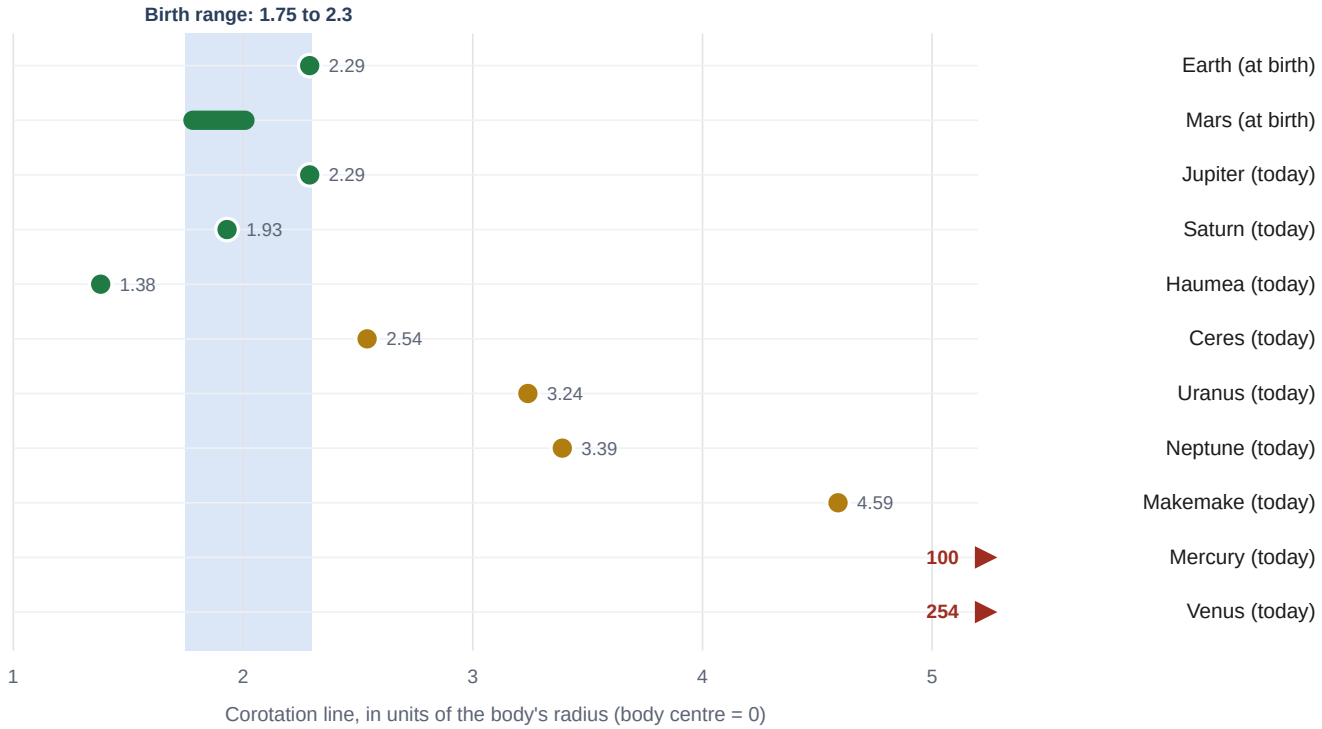


**Figure A.2.** The corotation line of each body, in units of the body's radius. The blue band is the pear and bulge range, measured on Earth and Mars. For Earth and Mars the value at the moment of birth is shown; for the other bodies, today's value. Jupiter and Saturn rotate today within the range. On Haumea the line is below the range: the tips of its long axis are already beyond it. On Mercury and Venus the line is very far away, and the arrow marks a value off the axis. *How to read:* each row is a body; the closer the point is to the band, the closer the body's spin is to the birth condition.

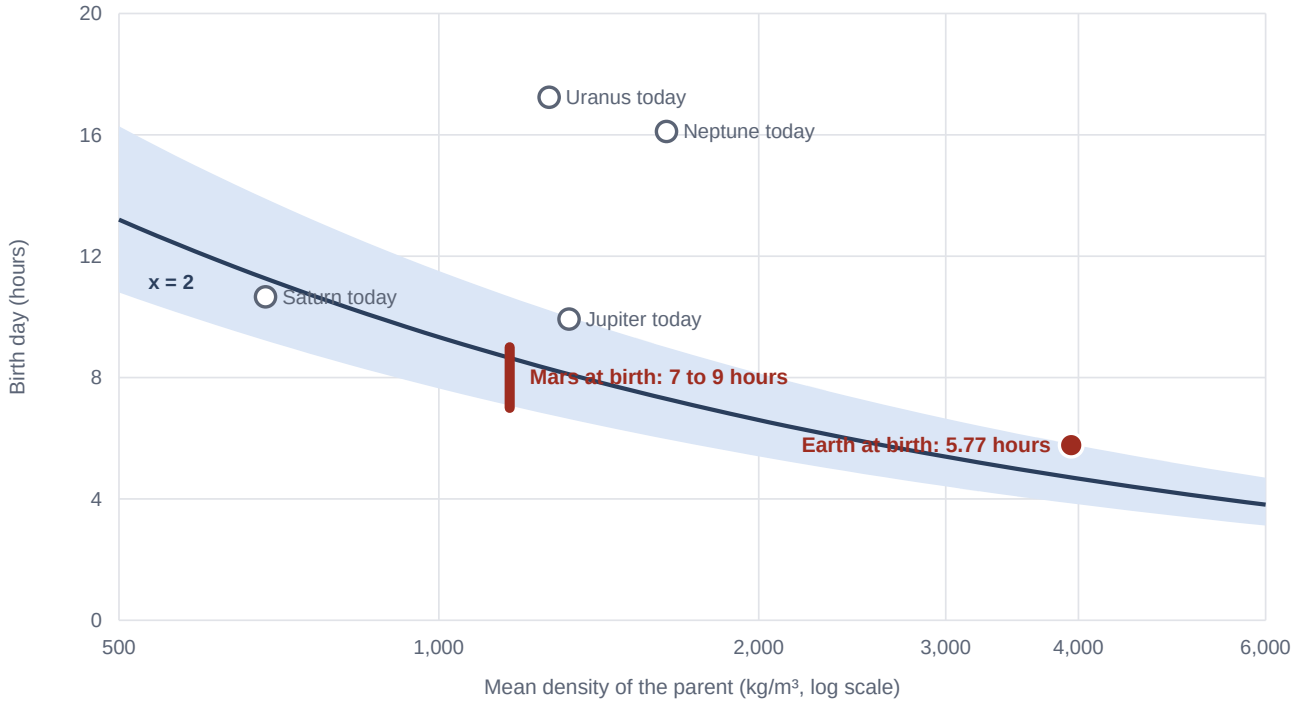



**Figure A.3.** The birth day as a function of the parent's density. The dark blue line is $x = 2$, and the band is the pear and bulge range, 1.75–2.3. The red points are Earth and Mars at the moment of birth, at their density then; both sit on the band. The open circles are the giants today, at today's density and spin. *How to read:* a planet below the band spins faster than required; a planet above it spins too slowly. Jupiter and Saturn lie on the band; Uranus and Neptune above it.

## A.4 Mechanism conditions

• **Mechanics of the pear and the bulge:** the formation of a structural elongation in the parent is a necessary condition for the birth of a moon. Fast axial spin pushes mass toward the geometric equator; the region in which most of the mass is concentrated is the most affected by the spin, and there the elongation begins to develop. This process continues as long as gravity at the equator is weakened by oblateness and a high spin rate. The elongation recedes when the parent slows, cools, and increases its viscosity as differentiation advances [Papers 3, 10].

• **Crossing the corotation line:** material that does not cross the corotation line collapses back onto the parent at the end of the process. Physical birth is defined at the exact moment when the centre of mass of the consolidating body crosses the corotation line; at this point, the centrifugal force combined with internal hydrostatic pressure tears the body away from the parent [Paper 3]. The remaining material left below the line is gathered back into the interior of the planet.

• **Dynamics of the structural race:** development of the pear structure carries angular momentum outward and slows the planet's spin. The slowing of the spin in turn moves the corotation line outward beyond the reach of the body. Birth requires a strict race condition: the radial growth rate of the pear must exceed the rate at which the corotation line recedes. Earth's Moon was released because its rate of construction met this condition; had the pear been more massive or built more slowly, the line would have moved beyond the tip of the structure and birth would have been blocked [Paper 2].

• **The structural birth window:** birth is restricted to a defined geological stage, set by the parent's degree of differentiation and not by chronological age. The window opens when the planet is hot, of low viscosity, not fully sorted and spinning fast (on Earth the process took place several hundred million years after the start of accretion). The window closes in advanced stages of differentiation, with the organisation of the core, the rise in viscosity, the escape of water and the rise in mean density. The length of the window in years varies from body to body: bodies that dried quickly (Mercury and Venus) closed the window early in chronological terms, whereas the gas and ice giants evolved slowly and kept a long birth window.

• **Duration of differentiation by size and heat:** The rate of differentiation is set by the internal heat of the parent body. In a body that is too small, the heat is insufficient, and differentiation does not begin or remains partial (Callisto, k = 0.355). In a medium body (Earth, Mars), differentiation proceeds quickly. In a giant body, heat acts in two ways that delay separation: it dissolves the heavy elements in the metallic hydrogen [Wilson & Militzer 2012], and it drives convection currents that stir the material faster than it can settle. Separation begins only when the body cools, the currents weaken and the solubility drops; once layers have formed, the layers themselves suppress convection, and differentiation continues at a very slow rate [Leconte & Chabrier 2012]. Hence the long birth window of the gas giants. The dilute core measured in Jupiter (Part G) and in Saturn [Mankovich & Fuller 2021] is differentiation that is halfway along its course, not differentiation that failed. The principle is also observed in the gas envelope: helium rain – the separation of helium from

hydrogen and its sinking inward – is more advanced in Saturn, the smaller and colder planet, than in Jupiter [Mankovich & Fortney 2020].

• **The hydrodynamic pump principle:** the sinking of dense materials into the planet's core pushes water and light silicate materials upward from the depths of the closed body. These materials concentrate at the exit openings – such as the volcanic structures on the bulge or the tip of the pear. These openings feed the birth of moons and rings, and through them mass carrying angular momentum is ejected outward. The pump is active as long as differentiation continues and the static crust is not thick enough to block its outlet. On Mars, the separation of metals into the core lasted only a few million years [Dauphas & Pourmand 2011], but structural differentiation and the hydrodynamic pump continued to operate until the end of the wet period, about 3.7–3.5 billion years ago, a time that matches the position of Phobos at the end of the ring stage.

• **The atmospheric cap principle:** the gas giants captured a hydrogen and helium envelope from the protoplanetary disc before it was dispersed by solar radiation [Haisch et al. 2001]. Capture of the gas envelope became possible only after the inner core body reached a critical mass [Pollack et al. 1996]. Because of their accretion distances, Jupiter ($5.2\ \mathrm{AU}$) accumulated the most massive envelope, Saturn ($9.5\ \mathrm{AU}$) a smaller envelope, whereas Uranus and Neptune ($19\ \mathrm{AU}$ and $30\ \mathrm{AU}$) captured a gas envelope amounting to only 10%–20% of their total mass. The gas envelope acts as a thermodynamic cap over the water: water (molecular weight 18) is heavier than the gas surrounding it (mean molecular weight about 2.3), condenses in the cold atmospheric layers and descends as precipitation into the body's interior (on Jupiter, water clouds are identified at a pressure of 5–7 bar [Li et al. 2020]).

### The mechanistic chain of the cap principle

Distance from the Sun → mass of gas accumulated → thickness of the atmospheric cap → blocking/allowing water escape from the line → change in angular momentum and in the position of the line

• **Jupiter and Saturn:** thick atmospheric cap $\rightarrow$ water escape prevented $\rightarrow$ the corotation line remains static at its original position.

• **Uranus and Neptune:** thin atmospheric cap $\rightarrow$ water escape through the corotation line $\rightarrow$ angular momentum carried outward $\rightarrow$ radial recession of the line.

### Modes of birth in the solar system

| Mode of birth | Mechanical mechanism | Properties of the resulting moon | Location in the solar system | Structural example |
|---|---|---|---|---|
| Built on the tip | Prolonged accumulation of material at the tip of the pear; detachment when the centre of mass crosses the line. | Dry, rotationally locked, of high mass relative to the parent. | Inner edge (dry zone, no free water for ejection). | Earth's Moon ($1.2\%$ of the parent's mass). The only body in this category. |
| Ejected | Volcano on a bulge; the pump injects viscous, wet silicate material under internal pressure. | Tiny mass relative to the parent ($0.01\% - 0.02\%$), rich in water/ice components. Irregular or round shape (depending on consolidation time). | Central zone (Mars to Neptune; massive, wet bodies). | Phobos and Deimos; the regular moons of the gas and ice giants. |
| Split | Spin beyond the hydrodynamic elongation threshold; stretching into a two-lobed peanut structure and tearing at the neck. | Very high mass relative to the parent; strict mutual tidal locking. | Outer edge (small, cold bodies with high spin rate). | Pluto and Charon ($89\%$ and $11\%$ of the system's mass respectively). |

### Geophysical distinction of ejected moons

**1. Reduced density:** in systems with moons of different sizes, the small moons are less dense than the large moons of the same system (for Jupiter: Amalthea $857\ \mathrm{kg/m^3}$ versus the Galilean moons $1,834 - 3,528\ \mathrm{kg/m^3}$; for Saturn: the ring moons $400 - 640\ \mathrm{kg/m^3}$ and Hyperion $544\ \mathrm{kg/m^3}$).

**2. Effect of distance from the Sun:** distance from the Sun lengthens the soft, wet stage of the material. As a result, distant, low-density moons are able to reach hydrostatic equilibrium and become round at small diameters (Mimas: diameter $396\ \mathrm{km}$, density $1,149\ \mathrm{kg/m^3}$; Miranda: diameter $470\ \mathrm{km}$, density $1,200\ \mathrm{kg/m^3}$).

**3. Structural rounding thresholds:** a body composed mainly of ice (Volatiles) reaches a rounded shape from a diameter of about $400\ \mathrm{km}$. A body composed of silicates requires a minimum diameter of about $600\ \mathrm{km}$ to become round.

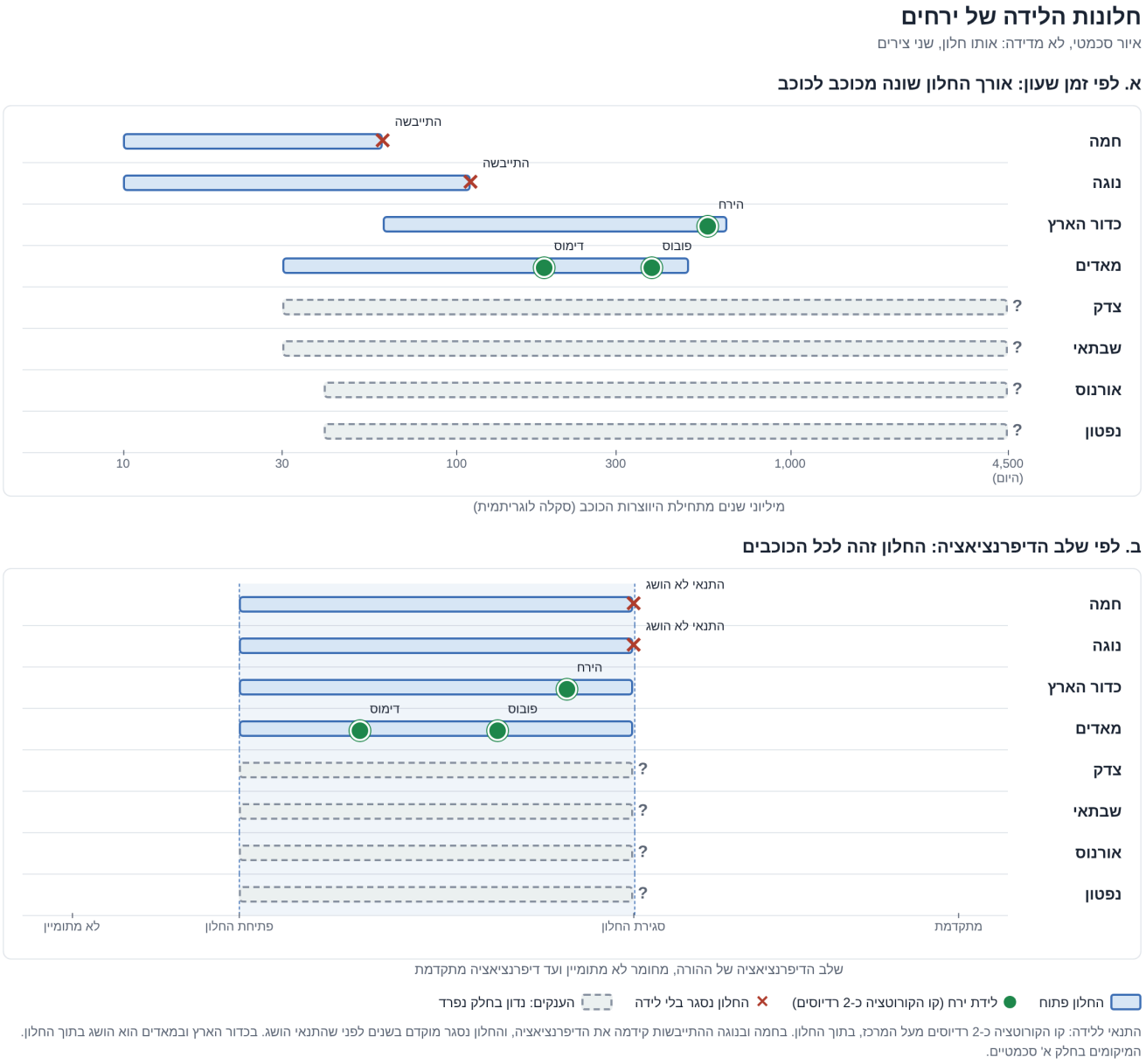



**Figure A.4.** The birth windows, shown schematically. *Top:* in clock time, the length of the window differs from planet to planet — on Mercury and Venus it closed early, because they dried out; on Earth and Mars birth took place within it; in the giants it is long. *Bottom:* by stage of differentiation, the window is the same for all planets, and the only difference is whether the spin condition was reached within it. The positions in the upper panel are illustrative only.

## A.5 Contracting inside, swelling outside: how differentiation brings the line closer

- **The paradox of global swelling:** the birth window is set solely by the stage of differentiation (A.4). However, the simplistic assumption that "the swelling of the planet brought the surface closer to the corotation line" does not hold mechanically. A body expanding uniformly without losing angular momentum would necessarily reduce its axial spin rate (like a skater on ice spreading her arms), which would cause a rapid radial

recession of the corotation line, at a rate higher than the rate at which the surface approaches.

• **The mechanism of volumetric variation:** the planet does not undergo homogeneous expansion. During structural differentiation, a massive contraction of the inner core takes place in parallel with the swelling of the outer layers, the central contraction being the mechanism that drives the ejection of the light materials. The dense components, which make up most of the body's mass, sink to the centre and accelerate the axial spin (like a skater drawing in her arms). At the same time, water and volatile components are pushed upward and swell the outer layer. This layer has a large volume but negligible mass, and it therefore increases the overall outer radius with almost no slowing of the axial spin. The dynamic combination of accelerated spin and the rise of the geometric surface brings about the meeting with the corotation line.

• **Mathematical development:** the angular momentum of the parent is defined by:

$$L = kMR^2\Omega$$

where $k$ is the moment-of-inertia factor (for a body of uniform density $k = 0.4$, and its value decreases as mass concentrates toward the centre). Under conservation of angular momentum, the spin rate satisfies:

$$\Omega \propto \frac{1}{kR^2}$$

Substituting this relation into the equation for the position of the corotation line (A.3) yields the structural relation:

$$x \propto (k^2 R)^{1/3}$$

While the swelling of the outer layer increases the radius $R$, the contraction of the inner core reduces the factor $k$. Since $k$ appears squared, the effect of the geodynamic contraction is dominant and drives the approach of the line.

**Effect of a change in moment of inertia on the position of the line ($x$)**

The values are calculated from a starting point at which the line stands at 2.1 radii, for an outer radius expanded by a factor of 1.3 ($R_{\text{after}} = 1.3R_{\text{before}}$):

| Geodynamic mechanism in the parent | Moment-of-inertia factor ($k$ before $\rightarrow$ after) | Final outer radius | Final position of the corotation line ($x$) |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Homogeneous swelling only (no internal change in density) | $0.4 \rightarrow 0.4$ | ×1.3 | 2.29 (line recedes) |
| Slight inner contraction and outer swelling | $0.4 \rightarrow 0.3$ | ×1.3 | 1.89 |
| Moderate inner contraction and outer swelling | $0.4 \rightarrow 0.25$ | ×1.3 | 1.68 |
| Strong inner contraction and outer swelling | $0.4 \rightarrow 0.2$ | ×1.3 | 1.44 |
| Very strong inner contraction and outer swelling | $0.4 \rightarrow 0.15$ | ×1.3 | 1.19 |

For geophysical comparison, Saturn's moment-of-inertia factor today is about 0.21. Passing from the structure of a uniform body to a structure with Saturn's parameters brings the line to about $1.4R$, and combined with expansion of the outer radius by a factor of 1.3 – to about $1.5R$ .

• **Geodynamic conclusion:** the differentiation process does not amount merely to raising silicate material to the ejection opening; it is the physical mechanism that brings the corotation line closer to the surface. The sinking of dense materials spins the body up, while the ejection of volatiles expands the outer envelope, until a mechanical meeting between the surface and the line. This is the physical reason why the birth window is defined by the structural stage of differentiation and not by the chronological time variable.

**Geodynamic application of the contraction-and-swelling principle**

• **The Jupiter system:** defines the mechanism of the chronological order of birth. As differentiation advances, the corotation line moves inward over time. As a result, moons formed at later stages consolidated on lines closer to the parent: Callisto first (on the most distant line), and Io last (on the closest line). The order of radial distance matches exactly the chronological order of birth. This process continues today, owing to the steady sinking of the dilute core (Part G).

• **The Pluto–Charon system:** structural differentiation pushed the united body beyond the hydrodynamic separation threshold (Rotational fission). At a fixed value of angular momentum, for a uniform homogeneous body ($k = 0.4$) the axial spin period is 2.51 hours, a value lying just short of the separation threshold (2.44 hours). As differentiation advanced and the moment-of-inertia factor fell to $k = 0.31$, the axial spin accelerated to 1.94 hours – a

value that crosses the split threshold and causes the neck of the peanut structure to tear (K.1).

• **Haumea (Haumea):** structural prediction condition: continued inner contraction of the dwarf planet's body will accelerate its axial spin rate and bring the body closer to a hydrodynamic separation threshold (K.2).

• **The Uranus and Neptune systems:** since the birth stage, an inner contraction of the core has taken place, which by itself should have brought the corotation line inward. In practice, a recession of the line from $2R$ to a range of $3.2R - 3.4R$ is identified. Hence the rate of water escape carried angular momentum outward with a strength that overcame the effect of inner contraction; therefore, an escape value of 2%–3% of the planet's mass is only a lower bound (Part I).

• **Earth and Mars:** the birth window opened in both bodies during a stage of intense differentiation: the stage of iron-core formation on Earth, and the stage of hydrodynamic pump activity on Mars. At this stage the corotation line came as close as possible to the surface. The physical principle does not change the calculated numerical values, but validates the geological timing.

Methodological rule for protocol check 3 (Part B): when calculating the dynamical reconstruction of the ancient spin rate, the exact values of the physical radius and of the moment-of-inertia factor ($k$) at the moment of birth must be used, and not the current parameters of the present era, so as not to omit the geodynamic effect described.

## A.6 Survival conditions

• **The dynamical Roche limit:** within the Roche limit, the tidal forces exerted by the parent exceed in strength the self-gravity component of a given mass. This condition entirely prevents the hydrostatic consolidation of new moons, and causes the mechanical break-up of any body that crosses the limit inward. The planetary ring systems of the gas and ice giants are all located within this dynamical region.

• **The role of planetary rings:** a ring system is a site of residence and continuous mechanical erosion, not a site of moon formation (accretion). Moons found today near ring systems represent bodies that migrated dynamically out of the ring's range (inward or outward). The planetary ring acts as an accompanying mechanism that straightens and circularises the eccentricity of the moon's orbit, and is able to drive a radial push outward, as was carried out for Deimos in the Mars system [Paper 10]. The primary mass-ejection mechanism is driven by the pear or bulge structure, not by the ring. In situations where water

components leak from the parent into the surroundings of the corotation line and are trapped within the Roche limit, the ring is a continuous dynamic mass flow, renewed as long as the ejection pump is active [Part H].

• **Hill radius and the influence of the Sun:** each planetary body has a closed gravitational region in which its gravitational field dominates over that of the Sun – the Hill radius ($r_H$), calculated as:

$$r_H = a\left(\frac{M}{3M_\odot}\right)^{1/3}$$

where $a$ is the semi-major axis of the planet's orbit, $M$ its mass, and $M_\odot$ the mass of the Sun. Close to the Sun, the Hill radius shrinks significantly. In addition, the axial tidal forces exerted by the Sun slow the planet's spin and move the corotation line radially outward. When the corotation line crosses the boundary of the Hill radius, every potential moon orbit lies below the line, which forces inward orbital migration and collapse onto the parent [Ward & Reid 1973; Burns 1973].

• Direction of tidal migration:

  • **Moon below the corotation line:** the orbital velocity is higher than the spin rate of the parent $\rightarrow$ the tidal bulge induced on the parent lags behind the moon and exerts a braking force $\rightarrow$ loss of orbital energy and radial migration inward.

  • **Moon above the corotation line:** the orbital velocity is lower than the spin rate of the parent $\rightarrow$ the tidal bulge induced on the parent leads the moon and accelerates it $\rightarrow$ gain of orbital energy and radial migration outward.

• **Tidal locking and internal structure:** a moon consolidating on the tip of the pear structure is released in a state of strict, full tidal locking (synchronisation in which the orbital period equals the spin period). Moons ejected from a volcano can become locked. The existence of an independent spin requires a dynamic, active core within the moon, which carries out its own differentiation and distributes the mass in a homogeneous, symmetric way.

## A.7 The required signatures

| Structural signature | Geophysical indicator and required finding |
|---|---|
| Parent material | Chemical composition and isotopic values close or identical to those of the parent, with no foreign mass components apart from secondary accretion of space dust. |

| | |
|---|---|
| Density as evidence | The moon's density reflects the exact composition of the material at the geological exit opening at the moment of birth, which testifies to the parent's stage of differentiation (A.9). |
| Structural sponginess | A moon ejected in a water-laden state undergoes rapid escape of volatiles (Outgassing), which leaves internal voids (Macro-porosity). The bulk density is lower than the mineral density of the constituent material. |
| Parent–moon density gradient | At the end of the process, the parent (the silicate/metallic part, without the gas envelope) is denser than its moons, because differentiation continues in the parent and stops in the moon. |
| Concentration of heavy elements | A moon formed at advanced stages of the parent's differentiation will contain a higher percentage of dense elements (iron and metals) and a defined core. |
| Dynamics on both sides of the line | Presence of small moons on both sides of the corotation line: those below it migrate inward, and those above it migrate outward. |
| Frozen morphology | Preservation of a geometric deformation (Fossil bulge) matching the fast spin and strong tidal forces of the ancient birth site, owing to rapid hardening. |
| Structural signatures on the parent | Morphological remnants of the pear or bulge structure: on Earth – the Pacific anomaly and the stripes [Paper 9]; on Mars – the Tharsis bulge (Tharsis) [Paper 10]. |

## A.8 The super-parameter

Definition: most of the hydrodynamic conditions converge into a single dimensionless structural parameter: the ratio between the radius of the corotation line and the outer physical radius of the parent:

$$x = \frac{a_c}{R}$$

Low mean density and axial oblateness increase the outer radius $R$; low surface gravity follows from an expanded radius $R$; and a high spin rate reduces the corotation radius $a_c$. Physical birth takes place within the differentiation window when the parameter $x$ reaches the defined mechanical ranges: $x \approx 1.75 - 2.3$ for a pear and a bulge, and $x \approx 0.85 - 1.5$ for an adjacent volcano and a hydrodynamic split. Earth and Mars met this condition at the moment of birth, while Jupiter and Saturn keep values of $x$ within this range in the present era as well.

## A.9 Stages of differentiation and the material at the opening

Structural differentiation is a continuous process in time. The composition of the material available at the ejection opening changes chronologically, and therefore moons formed in time

sequence from the same parent will display different densities and chemical compositions, which form a cross-section of the ejection opening at the moment of birth.

| Stage in the parent | Internal geodynamic mechanism | Composition of the material at the opening | Properties of the resulting body | Examples in the system |
|---|---|---|---|---|
| **0. Onset of differentiation:** initial stage, no deep sorting; an outer layer rich in primordial water that is ejected because of fast spin (centrifugal spray effect). | Water and volatile components from the outer layer. | Consolidation above the Roche limit; an icy body of very low density. | Iapetus, Hyperion (Hyperion). | |
| **1. Early stage:** mixed inner material, start of sinking of heavy components. | Mixed material representing the composition of the parent as a whole. | Massive body, density close to that of the parent, low internal differentiation. | Callisto, Titan, Charon. | |
| **2. Peak of the pump:** rapid sinking of the dense core; hydrodynamic push of water and light silicates upward. | Material enriched in water and light sialic components. | Water-rich, low-density body; post-birth escape of volatiles produces structural sponginess. | Phobos, Deimos, Tethys. | |
| **3. Final stage:** sorting almost complete; full escape of water or its confinement beneath the cap. | Dense, dry silicate residue. | Silicate moon, dense, free of water or ice. | Io (Io). | |

• **Test case – the Galilean system of Jupiter:** the four moons display a variation matching the sequence of stages, from the most distant to the closest (chronological order from earliest to latest):

**1. Callisto:** density $1,834\ \mathrm{kg/m^3}$. Low degree of differentiation, moment-of-inertia factor $k = 0.355$ [Anderson et al. 2001], which indicates a structure close to a uniform body (stage 1).

**2. Ganymede:** density $1,936\ \mathrm{kg/m^3}$. A sorted body with its own magnetic field [Kivelson et al. 1996] and a massive ice and water envelope.

**3. Europa:** density $3,013\ \mathrm{kg/m^3}$. A defined water envelope of about 100 km above a dense silicate body [Anderson et al. 1998].

**4. Io:** density $3,528\ \mathrm{kg/m^3}$. A dry, fully sorted body with a defined metallic core [Anderson et al. 1996] (stage 3).

- **The sponginess mechanism (Porosity):** the difference between the density of Phobos and Deimos ($1,860\ \mathrm{kg/m^3}$ and $1,470\ \mathrm{kg/m^3}$) and the density of Mars arises from ejection at stage 2. The hydrodynamic pump fed the opening with water-rich magma; with the ejection of the blocks and the rapid escape of the water into space, massive internal voids were left. The calculated porosity is about 30% for Phobos [Andert et al. 2010] and about 40% for Hyperion [Thomas et al. 2007].
- **Purity of the water and the stage of differentiation:** A distinction must be made between the purity of the water layer in the parent body and the material that reaches the vent. As differentiation advances, the water separates from the heavy material and rises into a distinct, cleaner layer. Slow freezing of the outer layer adds further cleaning: the growing ice rejects salts and mixtures into the liquid below it, so the ice is clean and the water beneath it is saltier, as found in the ice grains ejected from Enceladus [Postberg et al. 2009]. The cap determines whether this layer reaches the vent: when the water is retained, the ejected moon is icy and clean; when the water has escaped or been exhausted, the ejected moon is dry and dense (stage 3). Evidence: from the undifferentiated Callisto (k = 0.355) through Ganymede (k = 0.311 [Anderson, Lau et al. 1996]) to Europa, the water layer separates from the dense body and the surface becomes brighter (albedo 0.22, 0.43, 0.67). Consistency: the surfaces of Saturn's moons and its rings are the cleanest ice measured in the Solar System – Tethys density 984 kg/m³, Enceladus albedo 1.38 [Verbiscer et al. 2007], and the rings above 95% water ice [Cuzzi et al. 2010]. This finding also depends on the distance from the Sun, and it is therefore presented as a consistency, not as proof.

**Comparative analysis: moon density without the planetary gas envelope**

Since moons form from the parent's inner body and not from its atmospheric gas envelope (A.2), their density must be compared with the calculated inner component alone. Without a correction for gravitational compression, the mean density is translated into two basic components: a silicate-metallic part ($\rho_s = 3,500\ \mathrm{kg/m^3}$) and an icy part ($\rho_i = 950\ \mathrm{kg/m^3}$), according to the mixing equation:

$$\frac{1}{\rho} = \frac{f}{3500} + \frac{1-f}{950}$$

where $f$ represents the silicate mass fraction.

| Planetary system | Distance from the Sun | Atmospheric gas mass fraction | Mean density of the moons ($\mathrm{kg/m^3}$) | Calculated silicate mass fraction ($f$) |
|---|---|---|---|---|
| Earth | $1\ \mathrm{AU}$ | — | 3,344 | 0.98 |
| Jupiter | $5.2\ \mathrm{AU}$ | 90% – 97% | 2,578 | 0.83 |
| Saturn | $9.5\ \mathrm{AU}$ | 70% – 85% | 1,346 | 0.36 |
| Uranus | $19\ \mathrm{AU}$ | 10% – 20% | 1,505 | 0.49 |
| Pluto–Charon | $39.5\ \mathrm{AU}$ | — | 1,702 – 1,854 | 0.61 – 0.67 |

In massive moons (Ganymede, Titan) the effect of gravitational self-compression raises the measured density, but does not change the chemical trend.

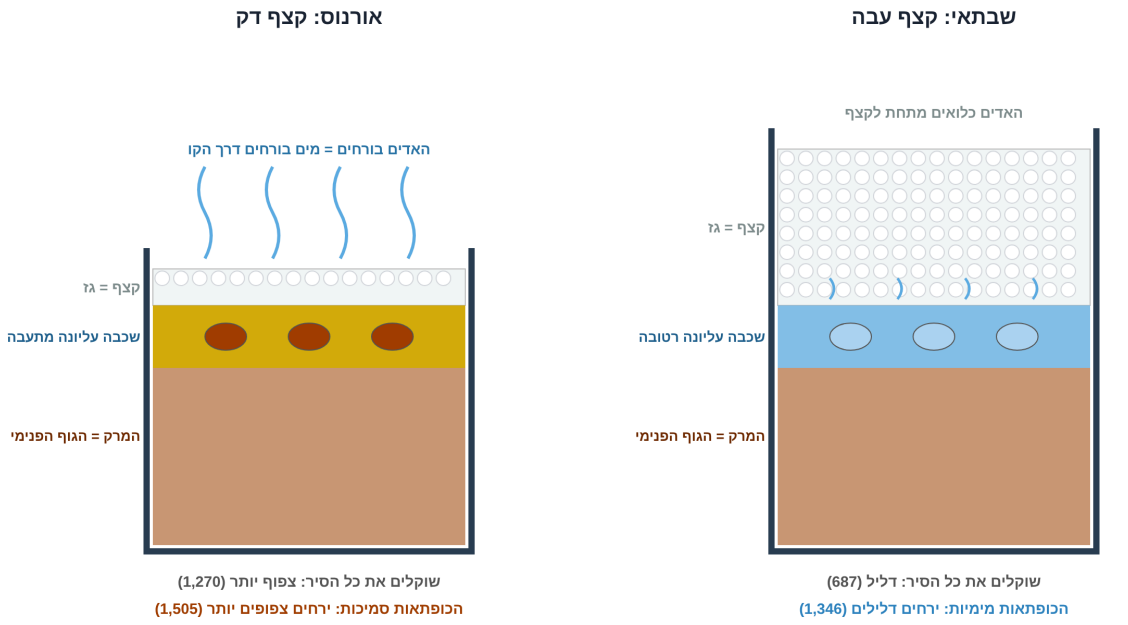
- **Integration of the early stage:** the massive moons ejected at the initial stage of differentiation (stage 1) converge to a narrow, uniform range of silicate mass fractions of $f \approx 0.6 - 0.7$: Callisto, Ganymede, Titan, Titania and Oberon, and alongside them the Pluto–Charon system.

**The break in the trend in the Uranus system: the two-pot model**

The mean density of the moons decreases radially from the Sun as far as the Saturn system, but rises again in the Uranus system. The anomaly is explained by a uniform thermodynamic model based on the thickness of the atmospheric cap (Figure A.5):

## בעיה 2: שני סירים של מרק – למה ירחי שבתאי דלילים מירחי אורנוס

הכופתאות (הירחים) נשלות מהשכבה העליונה של המרק. הן אינן עשויות מקצף. הקצף רק קובע אם האדים בורחים, ולכן כמה מים נשארים בשכבה העליונה.




| בסיר | בכוכב |
|---|---|
| קצף | מעטה הגז – המכסה |
| המרק | הגוף הפנימי (סיליקט, מתכת, מים) |
| אדים שבורחים | מים שבורחים דרך קו הקורוטציה |
| כופתאות מהשכבה העליונה | ירחים מהפתח |
| שקילת הסיר כולו | הצפיפות הממוצעת של הכוכב, כולל הגז |

הקצף עושה שני דברים בבת אחת: הוא מוריד את הממוצע כשמשקלים את כל הסיר, והוא כולא את האדים. לכן נראה ש"סיר דליל נותן כופתאות דלילות" – אבל שניהם תוצאה של אותו קצף. ההשוואה הנכונה היא בין הכופתאות לבין השכבה העליונה של המרק, לא לבין הסיר כולו.

**Figure A.5.** Two pots of soup. The foam is the gas envelope, the soup is the inner body, the steam is water escaping through the corotation line, and the dumplings are the moons, scooped from the top layer. On the left Uranus, with thin foam: the steam escapes and the top layer thickens. On the right Saturn, with thick foam: the steam is trapped and the top layer stays wet. *How to read:* weighing the whole pot (upper row beneath each pot) is affected by the foam; the dumplings (lower row) are affected only by the top layer.

- **The Saturn pot (thick cap):** massive gas envelope (70%–85% of the global mass) $\rightarrow$ full confinement of vapour and volatiles beneath the cap $\rightarrow$ the upper inner layer stays water-saturated and of low viscosity $\rightarrow$ the moons ejected from it have low density and a massive ice content (Tethys, Iapetus).
- **The Uranus pot (thin cap):** thin gas envelope (10%–20% of the global mass) $\rightarrow$ water and volatiles escaped freely through the corotation line into space $\rightarrow$ thickening and hardening of the parent's upper ejection layer $\rightarrow$ the ejected moons consolidate at a higher density ($f = 0.49$).

The proper geophysical comparison must be made between the density of the moons and the parent's upper ejection layer, not the global mean density of the planet as a whole. The water-

escape mechanism that dried the ejection layer on Uranus is the dynamical mechanism that moved its corotation line outward [Part I].

### A.10 The moon law

All the dynamical conditions, thermodynamic principles and geophysical signatures merge into a unifying physical law. A moon consolidates from the parent's material on the corotation line, within a window of opportunity defined by the stage of planetary differentiation. The full set of the moon's physical properties is set at three distinct points in time: before birth, at the moment of birth, and in the dynamical stage that follows.

A. Preconditions for birth

**1. The mechanical position criterion:** birth depends on the corotation line standing at the radial position matching the hydrodynamic mode of construction: about 2 radii ($1.75R - 2.3R$) for a pear and bulge structure, in the range $1.1R - 1.5R$ for a volcanic mechanism without a bulge, and about 1 radius for a full split. Since the position of the line in radius units is set solely by the physical ratio between spin rate and mean density ($\frac{\Omega^2}{\pi G\rho}$), a uniform ruler of thresholds applies to all celestial bodies.

**2. The birth-window criterion:** the time at which the birth window opens and closes is dictated solely by the parent's internal stage of differentiation, not by a static chronological measure of years.

**3. The hydrodynamic pump mechanism:** the differentiation process pushes water and light silicate components from the depths of the planet to the geological ejection opening. At the same time, contraction of the dense core accelerates the axial spin, while the ejection of volatiles swells the outer envelope; this combination mechanically brings the corotation line closer to the surface. Birth is functional as long as the pump is active and the outlet is open.

**4. The Sun as a thermodynamic clock (the arc of births):** birth requires active differentiation, a time window, a surface that is not sealed, and a subsurface layer of low viscosity. These factors depend on the radial distance from the Sun and produce an arc function:

- **At the inner edge (close to the Sun):** the parent undergoes rapid thermal drying that blocks the birth window at an early stage. Mercury and Venus did not give birth to moons; Earth gave birth to a single moon only because its position allowed prolonged accumulation of material on the tip of a pear.

• **In the central zone:** moving away from the Sun, the parents remain of low density, rich in water, and eject a larger number of moons, up to the peak in the Saturn system.

• **At the outer edge (beyond Saturn):** low temperatures cause the surface to freeze completely, and a rigid, frozen crust reduces the rate of moon ejection [L.4].

B. The mechanical mode of birth

**5. Radial lifting mechanism:** the planet's material is raised to the corotation line by means of a global pear structure, a bulge, or the cutting through of the whole body. Where the corotation line is close to the surface ($x \approx 1.1 - 1.5$), point volcanic activity suffices for ejection.

6. Morphological division by position in the system:

• **Built:** occurs at the inner edge only (Earth's Moon, a single case).

• **Ejected:** occurs in the central zone, from Mars to Neptune.

• **Split:** occurs at the outer edge, for small, cold bodies spinning beyond the break-up threshold (Pluto and Charon).

**7. Dependence of composition on the stage of differentiation:** the composition of the ejected mass is dictated by the geodynamic state of the parent:

• **At stage 0 (onset of the process, wet parent):** a mass of water is ejected from the outer layer by the centrifugal spray effect, and consolidates into a low-density icy moon only if it is located above the Roche limit.

• **At stage 1 (early):** mixed material representing the overall composition of the parent is ejected.

• **At stage 2 (peak of the pump):** material saturated with water and volatiles is ejected, leaving a moon of high structural sponginess after the gases escape.

• **At stage 3 (end):** a dense, dry silicate residue is ejected.

C. Structural properties of the moon

**8. Density:** reflects exactly the composition of the material at the ejection opening at the moment of birth. This value is compared with the parent's inner component, subtracting the volume of the atmospheric gas envelope. The thickness of the atmospheric cap sets the water content remaining in the parent's upper layer, and hence the density of the moon. At the end of the process, the parent is always denser than its moons (in comparison with its inner body, without the gas envelope).

**9. Sponginess (Macro-porosity):** rapid escape of water and volatile components after the ejection stage leaves massive internal voids in the body of the moon.

**10. Geometric morphology:** the dimensions of the mass and the length of the liquid/soft stage determine whether the body will reach hydrostatic equilibrium and become round.

**11. Spin dynamics:** moons formed by construction or splitting are born in a state of full tidal locking. Ejected moons can become locked. The existence of an independent spin requires an active, sorted core within the moon, which distributes the mass symmetrically.

**12. Primordial orbital parameters:** the moon moves in the direction of the parent's axial spin, in the plane of its geometric equator, and on an orbit with eccentricity close to zero (almost circular).

D. The post-birth dynamical evolution stage

**13. Radial recession of the corotation line:** the escape of water and volatiles through the region of the line carries essential angular momentum out of the system. As a result, the parent's spin rate decreases, and the corotation line is pushed radially outward at a declining rate (fast at first and slower later). Moons left below the migrating corotation line collapse back onto the parent; moons located above the line undergo tidal migration outward.

**14. Integration of the atmospheric cap:** a thick gas envelope traps the water inside and keeps a corotation line that hardly moves. A thin gas envelope allows free escape of water, which drives the recession of the line.

**15. Interaction with ring systems:** planetary rings are not a site of moon accretion, but a mechanical erosion mechanism that breaks them down and polishes them. The ring is able to accompany a moon, circularise its orbit and push it radially outward. Under continuous leakage, the ring is a constant mass flow that returns to the parent as precipitation ("ring rain").

**16. The feedback effect of the moon's mass:** a moon with a massive mass ratio relative to the parent causes a marked slowing of the parent's spin and a massive displacement of the corotation line (as in the Earth–Moon system). A tiny moon does not affect the parent's spin, and any displacement of the line arises solely from the escape of volatile mass (Mars and the gas giants).

## PART B The research protocol

Each planetary body in the solar system is examined systematically on the basis of 13 fixed checks applied in a strict chronological sequence, in order to carry out a parameter-based comparative analysis.

| # | Protocol check | Measured and calculated parameters |
|---|---|---|

| | | |
|---|---|---|
| 1 | Global basic data | Mass, physical radius, mean density, axial spin period (day), distance from the Sun, axial tilt. |
| 2 | The system's moon inventory | Catalogue and full separation between regular moons (Regular) and captured moons (Irregular). For each regular moon: mass, density, semi-major axis, orbital inclination, and direction of orbit. |
| 3 | Dynamical reconstruction of the primordial spin | Computational return of the moons' orbital momentum to the parent's axial spin, in order to determine the original spin rate before birth. |
| 4 | Validation of the birth law | Calculation of the theoretical birth day derived from the mass density at that time, and its comparison with the reconstructed spin rate and with the present value. |
| 5 | Analysis of water and volatile components | Determination of the chemical composition and thermodynamic history of the water in the parent during the estimated birth stage. |
| 6 | Present degree of differentiation | Measurement of the exact moment-of-inertia factor ($k = C/MR^2$): a value of 0.4 indicates uniform homogeneous density, and the value decreases as mass is concentrated in the core. |
| 7 | Definition of the birth window | Determination of the opening and closing points of the window according to the structural stage of differentiation, and their translation into an absolute measure of years from the scientific literature. |
| 8 | Tidal migration path | Calculation of the mechanical time required to move the moon from the position of the primordial corotation line to its present semi-major axis. |
| 9 | Dynamical survival parameters | Radial mapping of the Roche limit, the Hill radius, the strength of solar tides, and determination of the direction of tidal migration of the moons. |
| 10 | Morphological traces on the parent | Identification of structural remnants, gravity anomalies or geometric bulges that indicate the past existence of pear or bulge structures. |
| 11 | Structural traces on the moons | Analysis of chemical composition, density, presence of a core, tidal-locking status, and preservation of a frozen geometric deformation (Fossil bulge). |
| 12 | Analysis of ring systems | Mapping of their presence and composition, and definition of their mechanical evidence regarding the planet's stage of thermodynamic evolution. |
| 13 | Exceptions | Definition of the specific parameters in which the body does not fit the mechanism, and physical analysis of the causes of the deviation. |

## Methodology of check 3: reconstruction of the primordial spin

The angular momentum of the parent's spin is defined by $L_{\text{spin}} = CMR^2\omega$, and the orbital momentum of a moon on a circular orbit is defined by $L_{\text{orbit}} = m\sqrt{GMa}$ . The sum of these values is returned computationally to the parent's spin in order to reconstruct the original axial spin period.

The check is applied by substituting the exact values of the physical radius and the moment-of-inertia factor at the birth stage [A.5], since using the current parameters of the present era omits the geodynamic effect of inner contraction and outer swelling:

$$CMR^2\omega_{\text{then}} + m\sqrt{GMa_c} = CMR^2\omega_{\text{today}} + m\sqrt{GMa_{\text{today}}} + \Delta L_{\text{escaped}}$$

**where the position of the corotation line at the moment of birth is given by:**

$$a_c = \left(\frac{GM}{\omega_{\text{then}}^2}\right)^{1/3}$$

The parameters substituted in the equation are the parent's mass (derived from its volume and density), its physical radius, the moment-of-inertia factor, the present spin rate, the moon's mass and its present semi-major axis.

In the absence of an external momentum-loss component ($\Delta L_{\text{escaped}} = 0$), the equation reconstructs for the Earth–Moon system a corotation line at $2.29R$ for a primordial day of about 4.9 hours (a value close to the 5.77 hours defined in the series; the difference arises from changes in radius, in internal structure and in momentum loss due to solar tidal forces). In bodies where the mass of the moons is tiny relative to the parent, their orbital momentum does not affect the position of the line (the momentum of Titania is 0.4% of that of Uranus, the momentum of Titan is 1.4% of that of Saturn, and the momentum of Phobos is entirely negligible compared with that of Mars). In these systems, the present semi-major axis of the moon does not indicate the position of the primordial line, and any dynamical displacement of the line arises solely from the escaping volatile-mass component ($\Delta L_{\text{escaped}}$).

**Analysis of the reconstruction of the corotation line (simple form, without $\Delta L_{\text{escaped}}$)**

| Planetary system | Position of the line today ($R$) | Reconstructed primordial position of the line ($R$) | Fraction of angular momentum held by the moon |
|---|---|---|---|
| Earth and the Moon | 6.64 | 2.29 (day of about 4.9 hours) | 84% |

| | | | |
|---|---|---|---|
| Mars and Phobos | 6.03 | 6.03 | Negligible |
| Jupiter and Ganymede | 2.29 | 2.28 | 0.4% |
| Saturn and Titan | 1.92 | 1.90 | 1.4% |
| Uranus and Titania | 3.24 | 3.23 | 0.4% |

### Methodology of check 8: rate of tidal migration

The rate of radial displacement of the moon's semi-major axis as a function of time is defined by the differential equation:

$$\frac{da}{dt} = \frac{3k_2}{Q}\frac{m}{M}\sqrt{\frac{GM}{a^3}}\left(\frac{R}{a}\right)^5 a$$

The compound parameter $\frac{k_2}{Q}$ (made up of the Love number $k_2$ and the tidal quality factor $Q$) defines the parent's response to tidal deformation (its mechanical softness). Integrating the equation over time gives the length of time required to move the moon from the corotation line to its present position. The values of $\frac{k_2}{Q}$ are derived from direct empirical measurements [Lainey et al. 2009; 2017]. Since these values change over geological history, check 8 is not a tool for absolute dating, but a tool for testing mechanical consistency: determining whether the moon could have reached its present position within the given time window, or whether a secondary dynamical force is required to complete the displacement.

## PART C Mercury

Mercury is the body closest to the Sun and the smallest of the planets of the solar system. The body is moonless. According to the uniform mechanism, this finding matches the model's predictions exactly, owing to the simultaneous absence of several critical preconditions.

### Analysis of the 13 protocol checks – Mercury

| # | Protocol check | Findings and numerical values |
|---|---|---|
| 1 | Global basic data | Mass: $3.30 \times 10^{23}$ kg; physical radius: $2,440$ km; mean density: $5,427$ kg/m$^3$. Axial spin period (day): 58.6 Earth days (3:2 spin–orbit resonance). Axial tilt: close to $0^\circ$ (vertical axis). |

| | | |
|---|---|---|
| 2 | The system's moon inventory | Complete absence of regular or captured moons. |
| 3 | Dynamical reconstruction of the primordial spin | Initial value unknown. Solar tidal forces braked Mercury's axial spin until it reached the present resonant state, thereby completely erasing the primordial dynamical signature. |
| 4 | Validation of the birth law | On the basis of the present density ($5,427\ \mathrm{kg/m^3}$), the birth law defines a theoretical birth day of 3.3–4.9 hours. A body as dense as Mercury requires an extreme spin rate to reach the birth threshold. |
| 5 | Analysis of water and volatile components | Mercury is defined as a completely dry body with no history of liquid water. The MESSENGER mission identified volatile elements (potassium, sulphur and sodium) on the surface [Peplowski et al. 2011], and deposits of water ice were also located in thermal cold traps within craters at the poles [Lawrence et al. 2013]; these components are not evidence of internal liquid water, and the ice is preserved only in permanently shadowed regions. |
| 6 | Present degree of differentiation | A structurally advanced sorted body: a massive iron core amounting to about 85% of the total physical radius. |
| 7 | Definition of the birth window | Early closing of the birth window. The rapid drying of the material combined with massive sinking of the core closed the geological window before the spin conditions were reached. |
| 8 | Tidal migration path | Not applicable (no moons). |
| 9 | Dynamical survival parameters | The present corotation line stands at a radius of $242,900\ \mathrm{km}$ – a value located outside the gravitational hold defined by the Hill radius ($220,600\ \mathrm{km}$). |
| 10 | Morphological traces on the parent | No structural signatures of a pear or bulge were found. |
| 11 | Structural traces on the moons | Not applicable. |
| 12 | Analysis of ring systems | Complete absence of a ring system. |
| 13 | Exceptions | The body is not an exception to the model. Mercury does not simultaneously meet the criteria of water content, density, an open geological window and orbital stability. |

## The mechanism blocking dynamical survival close to the Sun

The absence of moons around Mercury is explained by a strict mechanical chain arising from thermodynamic proximity to the Sun:

Proximity to the Sun's gravitational field → reduction of the Hill radius (220,600 km) → braking of axial spin by solar tides (58.6 days)

⇓

Radial displacement of the corotation line outside the Hill radius (242,900 km)

As a result of this displacement, any potential moon orbit within the planet's region of hold is necessarily located below the corotation line. This dynamical configuration activates a negative direction of tidal migration, which forces the body to migrate inward until mechanical collapse onto the surface of the parent [Ward & Reid 1973; Burns 1973]. Mercury dried and reached high density before an effective birth window opened, and even had a moon formed, its dynamical stability would have been blocked.

## PART D Venus

Venus has structural parameters closest to those of Earth: its mass is about 82% of Earth's mass, and its mean density is almost identical. The body is moonless.

### Analysis of the 13 protocol checks – Venus

| # | Protocol check | Findings and numerical values |
|---|---|---|
| 1 | Global basic data | Physical radius: $6,052\ \mathrm{km}$; mean density: $5,243\ \mathrm{kg/m^3}$. Axial spin period (day): 243 Earth days, in the reverse direction (Retrograde). Axial tilt: $177^\circ$. |
| 2 | The system's moon inventory | Complete absence of regular or captured moons. |
| 3 | Dynamical reconstruction of the primordial spin | Value unknown. The dynamical combination of solar tidal forces and the coupling torque of the dense atmosphere can bring Venus to its present retrograde spin from any dynamical initial conditions [Correia & Laskar 2001]; therefore, it is possible that at early stages the spin rate was high. |
| 4 | Validation of the birth law | For the present density, the birth law requires a day of 3.3–5.0 hours; under the assumption of an expanded primordial radius (similar to Earth's parameters), a day of about 5.5 hours is required. |
| 5 | Analysis of water and volatile components | Venus underwent a massive loss of water. The deuterium-to-hydrogen isotopic ratio ($D/H$) in the upper atmosphere of Venus is 100 or more times higher than the value measured on Earth [Donahue et al. 1982]; this finding is an |

| | | |
|---|---|---|
| | | unambiguous geochemical signature of free thermal escape of light hydrogen and water into space. |
| 6 | Present degree of differentiation | The measured moment-of-inertia factor is $k \approx 0.34$ [Margot et al. 2021], a value matching Earth's degree of differentiation. The body is sorted. |
| 7 | Definition of the birth window | The escape of the water components raised the viscosity of the magma and the mean density of the body, thereby blocking and closing the birth window at an early stage. |
| 8 | Tidal migration path | Not applicable. |
| 9 | Dynamical survival parameters | The present corotation line stands at a distance of $1,536,500\ \mathrm{km}$, a value located outside the planet's Hill radius ($1,010,600\ \mathrm{km}$). |
| 10 | Morphological traces on the parent | There is a geometric signature to be examined: the "Aphrodite" highland (Aphrodite Terra) is a defined raised structure along the geometric equator. Verification is required as to whether this structure is a geodynamic remnant of a primordial bulge. |
| 11 | Structural traces on the moons | Not applicable. |
| 12 | Analysis of ring systems | Complete absence of a ring system. |
| 13 | Exceptions | The body is not an exception to the model. The absence of moons matches exactly the theoretical arc of births [L.4]. |

### Dynamics of birth blocking in the inner arc

The formation of a moon under the uniform mechanism requires the simultaneous existence of four physical conditions: active differentiation, a long time window, an open (not sealed) crust, and a subsurface layer of low viscosity.

Close to the Sun (Mercury and Venus), the rate of thermal loss is high and leads to rapid drying of the outer envelope. The thickening of the crust and the rise in viscosity of the material block the outlet of the hydrodynamic pump before the axial spin rate reaches the critical threshold defined in the birth law.

In addition, as in the Mercury system, Venus does not allow a stable moon orbit today: the corotation line ($1.53 \times 10^6\ \mathrm{km}$) extends beyond the boundary of the Hill radius ($1.01 \times 10^6\ \mathrm{km}$). Alternative hypotheses in the scientific literature have proposed that Venus gave birth to a primordial moon that was lost owing to the slowing of the axial spin [Alemi & Stevenson 2006].

A complete morphological exclusion of signs of a bulge or pear on Aphrodite Terra would finally validate that Venus never reached the mechanical birth stage.

## PART E Earth

Earth serves as the central calibration graph of the model. Applying the unified protocol to the system validates and confirms the numerical values and geophysical mechanisms developed in the series.

### Analysis of the 13 protocol checks – Earth

| # | Protocol check | Findings and numerical values |
|---|---|---|
| 1 | Global basic data | Present mean density: $5,513\ \mathrm{kg/m^3}$; present day: 23.9 hours. At the moment of birth: expanded primordial outer radius: $7,157\ \mathrm{km}$; density at that time: about $3,940\ \mathrm{kg/m^3}$; primordial day: 5.77 hours [Paper 2]. |
| 2 | The system's moon inventory | A single moon, amounting to 1.2% of the parent's mass. This is the highest mass ratio among the planets of the solar system, and the only body in the system built on the tip of a pear. Characterised as completely dry. |
| 3 | Dynamical reconstruction of the primordial spin | Full combination of the present momentum of the system into Earth's spin yields a day of 4.0 hours for the present radius, and 6.2 hours under the assumption of the expanded radius and unsorted material. The model value (5.77 hours) lies within these bounds; this value requires an angular momentum 8% higher than the system's present value, in agreement with the excess calculated in the series (11% for a day of 5.5 hours) [Paper 2]. |
| 4 | Validation of the birth law | The corotation line at the moment of mechanical birth stands at a radius of $16,400\ \mathrm{km}$, defined exactly as $2.29R$. |
| 5 | Analysis of water and volatile components | Primordial Earth was a wet body. The water was trapped and remained stored in the mantle and at the surface up to the present era. |
| 6 | Present degree of differentiation | Sorted. |
| 7 | Definition of the birth window | Opening and use of the birth window within several hundred million years of initial accretion. Defined chronological range: about 4.5 to 3.9 billion years before present. |
| 8 | Tidal migration path | Under the present tidal dissipation factor ($k_2/Q \approx 0.025$), the migration time required to move the Moon from the original corotation line to the present semi-major axis is 1.6 billion years. To match the absolute geological age (about 4 |

| | | |
|---|---|---|
| | | billion years), the mean dissipation factor in the past must have been about 2.5 times lower than the modern value, a finding that matches the oceanographic models in which the present ocean morphology produces a particularly high rate of energy dissipation [Bills & Ray 1999]. |
| 9 | Dynamical survival parameters | In order. Earth's Hill radius is about 1.5 million km, which allows full orbital stability of the Moon throughout the stages of migration. |
| 10 | Morphological traces on the parent | A structural remnant of the base of the geodynamic pear was found in the Pacific Ocean near the equator (Kiritimati), together with the array of structural stripes on the floor of the northern Pacific, which represent the leading side of the pear structure [Paper 9]. |
| 11 | Structural traces on the Moon | Oxygen isotopic values identical to those of Earth within the resolution of measurement [Young et al. 2016]. Presence of a tiny iron core (1%–2% of the mass). Synchronously tidally locked. Preservation of a frozen geometric deformation (Fossil bulge) showing a symmetric elongation along the Earth axis that exceeds the theoretical value obtained at its present distance [Garrick-Bethell et al. 2006]. |
| 12 | Analysis of ring systems | Complete absence of a ring system in the present era. |
| 13 | Exceptions | The body shows no deviations; the full set of conditions and signatures is fully validated. |

### Morphology of the components in the Moon's frozen shape

**In the shape of the Moon two mechanical components can be separated and examined each on its own (Part M):**

**1. The asymmetric (one-sided) component:** thick crust on the far side and thin crust on the side facing Earth, which displaces the geometric centre of figure from the centre of mass. This structure indicates sinking and accretional landing of returning mass mainly on the far hemisphere of the Moon during the detachment stage.

**2. The symmetric (two-sided) component:** a symmetric geometric elongation along the Earth axis (an elongated, egg-like structure). This deformation may have formed as a result of direct mechanical stretching on the tip of the pear structure – Earth exerted gravitational pull on the near hemisphere, while corotation forces pulled the far hemisphere outward, and the body underwent thermal hardening (morphological freezing) while under these tensile stresses.

## PART F Mars

Mars is the second test case of the model. In this system, the moons Phobos and Deimos were ejected through a volcanic feeder conduit that formed on the bulge, a mechanism that dictates their tiny dimensions.

### Analysis of the 13 protocol checks – Mars

| # | Protocol check | Findings and numerical values |
|---|---|---|
| 1 | Global basic data | Present mean density: $3,934\ \mathrm{kg/m^3}$; present day: 24.6 hours. At the moment of birth: primordial radius expanded by a factor of 1.5 ($5,085\ \mathrm{km}$); density at that time: about $1,170\ \mathrm{kg/m^3}$; primordial day: 7–9 hours [Paper 10]. |
| 2 | The system's moon inventory | Two regular moons: Phobos (density: $1,860\ \mathrm{kg/m^3}$) and Deimos (density: $1,470\ \mathrm{kg/m^3}$). The bodies move on almost perfectly circular orbits and in the plane of Mars's geometric equator. |
| 3 | Dynamical reconstruction of the primordial spin | The orbital momentum of Phobos and Deimos is negligible, about one part in a million of Mars's spin momentum. A Mars expanded by a factor of 1.5 that kept the present angular momentum would rotate with a period of 61 hours. For an axial spin of 7–9 hours at the moment of birth, an angular momentum 7 to 9 times higher than the modern value is required. Strict mechanical requirement: Mars lost 85%–88% of its original angular momentum; this loss was driven by the escape of volatile mass (water) and momentum through the ring system and the corotation line [Paper 10]. |
| 4 | Validation of the birth law | The corotation line at the moment of birth stands in the range $8,800 - 10,400\ \mathrm{km}$, defined as $1.74R - 2.05R$. |
| 5 | Analysis of water and volatile components | Young Mars was a water-saturated body (relative content higher than that of primordial Earth). The body underwent a massive loss of water that was ejected through the region of the corotation line. |
| 6 | Present degree of differentiation | Sorted. |
| 7 | Definition of the birth window | The iron core of Mars formed at a very early stage, within a few million years of accretion [Dauphas & Pourmand 2011]. The separation of metals is an initial stage of differentiation, which continued to develop and drove the hydrodynamic pump (A.4), which carried water and magma from the sealed interior through the volcanic system of the bulge to the ring. The window was blocked and closed with the weakening of differentiation, the thickening of the crust and the end of supply at the close of the wet period, about 3.7–3.5 billion years ago – a time that matches the position of Phobos at the end of the ring stage. |

| | | |
|---|---|---|
| 8 | Tidal migration path | The reconstruction of the orbital history is detailed below and documented in Figure F.1. |
| 9 | Dynamical survival parameters | In order. |
| 10 | Morphological traces on the parent | The Tharsis complex (Tharsis) and the sharpening bulge; the combination of volcanic structures at the tip of the geometric bulge is the main ejection site [Paper 10]. |
| 11 | Structural traces on the moons | Both bodies have a density significantly lower than that of Mars ($3,934\ \mathrm{kg/m^3}$). Ejection at stage 2 (peak of the pump, A.9) as water-saturated magmatic blocks caused massive internal voids (Macro-porosity) to form as the volatiles escaped. The porosity of Phobos is calculated at about 30% [Andert et al. 2010]. Both moons are tidally locked. |
| 12 | Analysis of ring systems | Complete absence of a ring system in the present era; the model requires a ring system in the past [Paper 10]. |
| 13 | Exceptions | The body is not an exception; it matches the full set of the model's predictions. |

### Dynamics on both sides of the line in the Mars system

The corotation line of Mars in the present era stands at a distance of $20,427\ \mathrm{km}$ ($6.03R$). The moon system matches fully the pattern of both sides of the line that recurs in the solar system (Part L):

- **Phobos:** semi-major axis: $9,376\ \mathrm{km}$. Located below the corotation line $\rightarrow$ negative direction of migration $\rightarrow$ the body is in continuous orbital descent toward the parent.
- **Deimos:** semi-major axis: $23,463\ \mathrm{km}$. Located above the corotation line $\rightarrow$ positive direction of migration $\rightarrow$ the body undergoes slow orbital recession.

### Reconstruction of the tidal migration path of Phobos

The orbital descent of Phobos is measured today directly and empirically [Jacobson 2010], and defines an expected collapse onto the surface within tens of millions of years [Black & Mittal 2015]. Running the migration equation backward in time (integration based on the present tidal dissipation values of Mars) establishes that about 3.5 billion years ago Phobos stood at a radial height of $16,300 - 18,700\ \mathrm{km}$.

This value matches exactly the bound calculated in the series for the end of the planetary ring stage ($16,000 - 19,000$ km) [Paper 10]: the ring system drove a dynamical push of Phobos outward in parallel with the motion of the corotation line; with the drying of the hydrodynamic pump and the disappearance of the ring, the pushing force ceased and the moon began its inward tidal collapse. The dynamical precision obtained on the basis of present-era data (Figure F.1) validates the orbital history without the need for secondary assumptions, the action of the ring being the mechanism that kept the eccentricity and orbital inclination at zero. Deimos, because of its great radial distance, experiences entirely negligible tidal forces and therefore keeps its original position from the end of the pushing stage of the primordial ring.

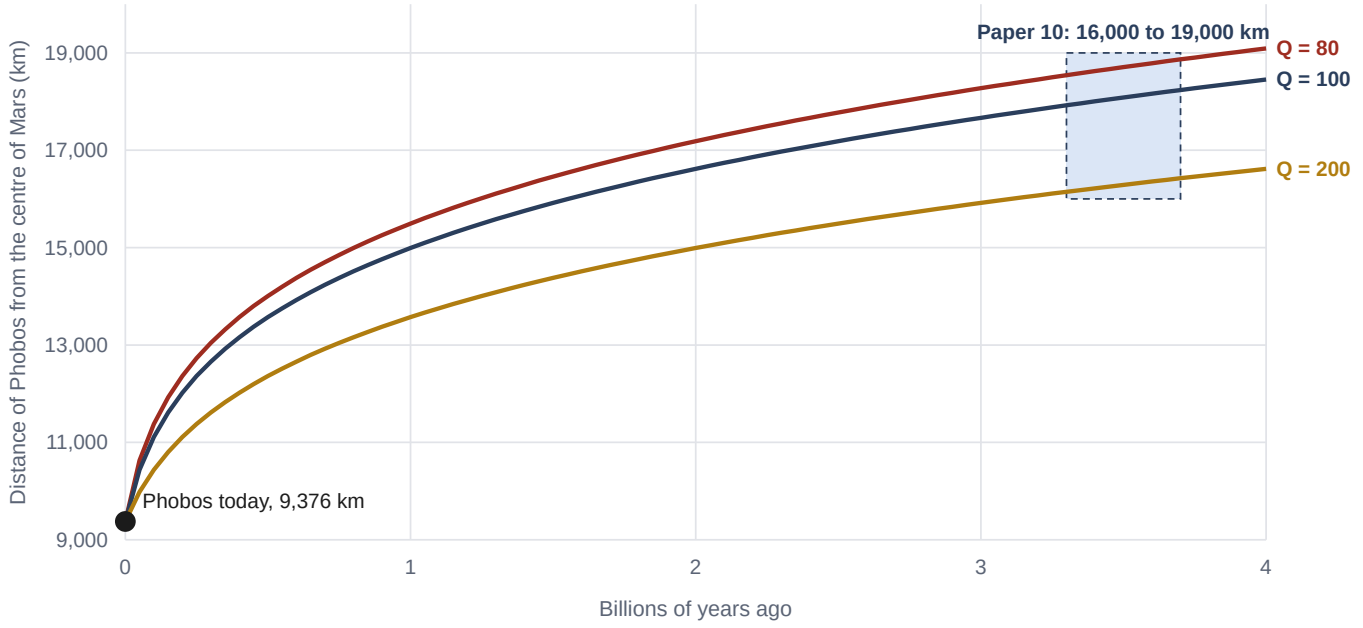



**Figure F.1.** The distance of Phobos from the centre of Mars in the past, from the backward calculation of tidal migration, for three values of Mars's $Q$. The black point is Phobos today. The dashed rectangle is the height that Paper 10 gives for the end of the ring period. *How to read:* each line is one possible history of Phobos; all the lines pass within or close to the rectangle about 3.5 billion years ago.

## PART G Jupiter

Jupiter is the most massive gas giant in the solar system. The system has eight regular moons: four small inner moons and the four massive Galilean moons. The dozens of outer moons, characterised by highly inclined or reverse-direction orbits (Retrograde), are defined as captured moons (Irregular) and are entirely excluded from the protocol checks.

### Analysis of the 13 protocol checks – Jupiter

| # | Protocol check | Findings and numerical values |
|---|---|---|
| 1 | Global basic data | Global mean density: $1,326\ \mathrm{kg/m^3}$; present day: 9.9 hours. The accretion of the body took place at a very early stage, within a few million years of the start of the formation of the solar system [Kruijer et al. 2017], but its structural and geodynamic evolution is particularly slow. |
| 2 | The system's moon inventory | Catalogue of the regular moons: four small inner moons (Metis, Adrastea, Amalthea, Thebe) and four massive Galilean moons (Io, Europa, Ganymede, Callisto). |
| 3 | Dynamical reconstruction of the primordial spin | The total orbital momentum of the moon system is only about 1% of Jupiter's axial spin momentum. Full computational return of this momentum reduces the axial day from 9.9 hours to only 9.8 hours. The finding indicates that Jupiter has undergone an entirely negligible slowing since the birth stage. |
| 4 | Validation of the birth law | Jupiter's corotation line in the present era stands at a distance of $160,000\ \mathrm{km}$ from its centre, defined exactly as $2.29R$ – a numerical value identical to the parameter calculated for Earth at the moment of the Moon's birth. |
| 5 | Analysis of water and volatile components | The body formed beyond the system's snow line (Snow line); as a result, the planet and its moon system are rich in water and water ice. |
| 6 | Present degree of differentiation | The findings of the Juno probe indicate a dilute, blurred core (Dilute/Fuzzy core), in which the dense elements are mixed with metallic hydrogen to a depth of about half the planetary radius [Wahl et al. 2017]. This finding is an unambiguous signature of structural differentiation that has not been completed. |
| 7 | Definition of the birth window | Open in principle (see details below). |
| 8 | Tidal migration path | Under the empirical dissipation factor measured for Jupiter ($k_2/Q \approx 1.1 \times 10^{-5}$ [Lainey et al. 2009]), the time required to move Io from the original corotation line to its present position is calculated at about 600 million years. By contrast, Europa, Ganymede and Callisto cannot reach their present positions through the tidal mechanism alone (integration yields a theoretical migration time of tens of billions of years). Strict mechanical requirement: a secondary dynamical mechanism pushing outward is required (similar to the push of Deimos on Mars). |
| 9 | Dynamical survival parameters | In order. |
| 10 | Morphological traces on the | The thick atmospheric gas envelope blocks direct morphological observation of the core. |

| | | |
|---|---|---|
| | parent | |
| 11 | Structural traces on the moons | Validation of the density gradient, the structure of the cores and the locking status (detailed below). |
| 12 | Analysis of ring systems | A system of thin dust rings, located in a radial range of $1.29R - 3.1R$, extending on both sides of the corotation line [Owen et al. 1979]. |
| 13 | Exceptions | The body is not an exception. The gradient in which Jupiter's density is lower than that of its moons is fully explained by isolating the gas-envelope component (A.2). |

### Mechanics of both sides of the line in the inner moons

Jupiter's system of small inner moons preserves exactly the pattern of both sides of the line defined in the model (Part L):

- Below the corotation line ($x < 2.29$): the moons Metis and Adrastea are located at $1.83R$ and $1.85R$ respectively; they have a negative direction of migration and are in inward orbital descent.
- Above the corotation line ($x > 2.29$): the moons Amalthea and Thebe are located at $2.59R$ and $3.17R$ respectively; they have a positive direction of migration and are in outward orbital recession.

### The chronological density gradient and the internal structure of the Galilean moons

The density of the Galilean moons shows a monotonic radial decrease as their distance from Jupiter increases (Figure G.1):

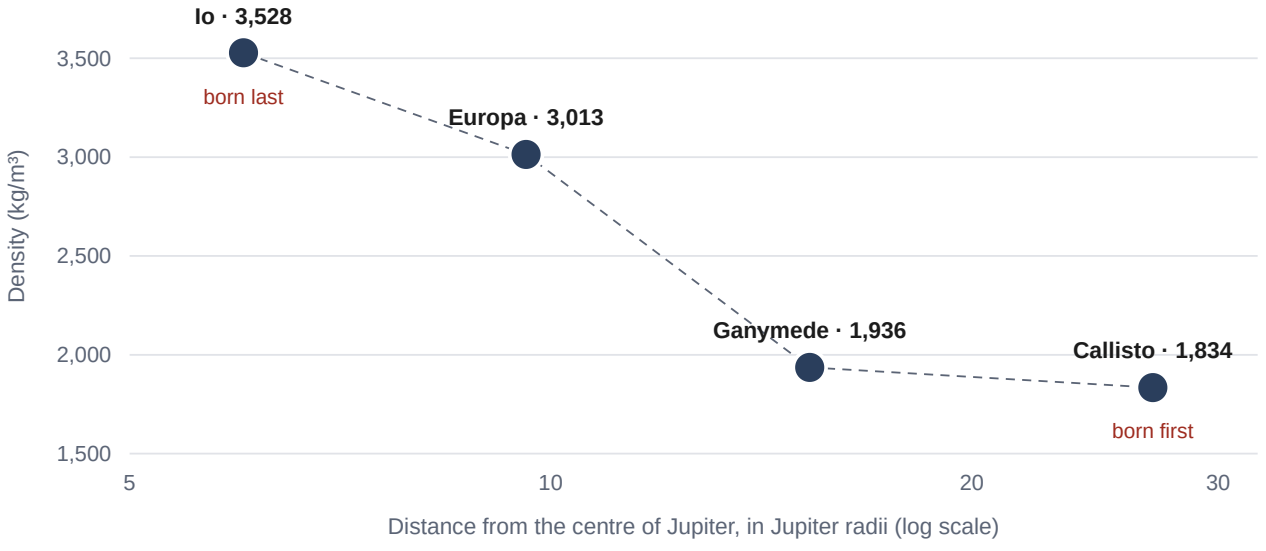


**Figure G.1.** The density of the four Galilean moons against their distance from Jupiter. Density decreases with distance. *How to read:* the density reflects the material at the opening at the moment of birth, and therefore the order from right to left is the order of birth and of the stages of differentiation — Callisto first, at its start; Io last, at its end.

Io (3,528 kg/m³) → Europa (3,013 kg/m³) → Ganymede (1,936 kg/m³) → Callisto (1,834 kg/m³)

This gradient validates the model's statements without the need for secondary assumptions, and indicates that the order of radial distance matches the chronological order of birth (from earliest to latest) through the inward motion of the line toward the surface (A.5):

**1. Callisto (born first):** ejected at stage 1 (onset of Jupiter's differentiation). Its moment-of-inertia data ($k = 0.355$ [Anderson et al. 2001]) indicate a mixed homogeneous structure close to a uniform body, as expected from ejection before the parent was sorted. Its surface is the oldest in the system and saturated with craters [Zahnle et al. 2003].

**2. Ganymede and Europa (intermediate stage):** ejected at stage 2 (peak of the hydrodynamic pump's activity). The bodies are structurally sorted and carry massive water and ice envelopes; Ganymede has an active liquid iron core that drives its own magnetic field [Kivelson et al. 1996], and Europa has a dense core and a water envelope of about 100 km [Anderson et al. 1998].

**3. Io (born last):** ejected at stage 3 (end of differentiation). The body is completely dry, dense, and contains a defined, massive metallic core [Anderson et al. 1996], as expected from ejection after the volatiles and water were trapped beneath the atmospheric cap. Its surface is the youngest in the system owing to continuous volcanic activity [Zahnle et al. 2003].

**Mode of birth and cap dynamics at present**

All of Jupiter's moons formed by a successive ejection mechanism (ejected), as evidenced by their tiny mass ratio of $0.008\%$ or less of Jupiter's mass (hundreds of times lower than the ratio of the Earth system).

Jupiter rotates even today within the theoretical birth range ($x = 2.29$), and its internal differentiation continues (sinking of the dilute core), which leaves the birth window mechanically open. However, the formation of new moons is completely blocked by the action of the atmospheric cap: the massive hydrogen and helium envelope (90%–97% of the planet's mass) acts as a strict thermodynamic barrier over the water. Internal water piled up by the pump condenses in the deep atmospheric layers at a pressure of 5–7 bar [Li et al. 2020] and collapses back as precipitation. Only a tiny dust component reaches the corotation line and

feeds the thin ring system; material that crosses the line inward collapses into the interior of the planet, and the absence of a mass supply prevents new bodies from consolidating.

## PART H Saturn

Saturn has the lowest mean density in the solar system and the most massive planetary ring system. The system has about twenty regular moons, with most of the system's mass (about 96%) concentrated in the single moon Titan. The dozens of outer moons are defined as captured moons (Irregular) and are not included in the protocol checks.

### Analysis of the 13 protocol checks – Saturn

| # | Protocol check | Findings and numerical values |
|---|---|---|
| 1 | Global basic data | Global mean density: $687\ \mathrm{kg/m^3}$ (less than the density of free water). Axial spin period (day): 10.56 hours [Mankovich et al. 2019]. Axial tilt: $27^\circ$. |
| 2 | The system's moon inventory | About twenty regular moons, dominated in mass by Titan, alongside five medium-sized moons and a group of tiny ring moons. |
| 3 | Dynamical reconstruction of the primordial spin | The total orbital momentum of the regular moons is about 1.4% of Saturn's axial spin momentum. Full computational return of this momentum to the parent's spin reduces the primordial axial day to only 10.4 hours. |
| 4 | Validation of the birth law | Saturn's corotation line in the present era stands at a distance of $111,600\ \mathrm{km}$ from its centre, defined exactly as $1.92R$ – a value located within the hydrodynamic birth range of the uniform model. |
| 5 | Analysis of water and volatile components | Saturn formed beyond the snow line, and its inner body is extremely rich in water and water-ice components. |
| 6 | Present degree of differentiation | The inner core (metals, silicates and ice) is screened beneath a massive hydrogen and helium envelope. Gravity-field data indicate that the core is not fully sorted. The moon Titan is also only partially sorted ($k \approx 0.34$ [Iess et al. 2010]). |
| 7 | Definition of the birth window | The birth window is open in principle in the present era (see the flow dynamics detailed below). |
| 8 | Tidal migration path | Saturn is very soft to tidal deformation ($k_2/Q \approx 1.6 \times 10^{-4}$ [Lainey et al. 2017]). The axial tidal force alone supplies the dynamical energy required to move Mimas, Enceladus, Tethys and Dione from the corotation line to their present positions within a short time of 0.6–2.3 billion years. Current measurements indicate that Titan is receding today at a rate faster than expected in classical |

| | | |
|---|---|---|
| | | analytical models [Lainey et al. 2020], a finding that validates that it was significantly closer to the parent in the past. |
| 9 | Dynamical survival parameters | In order. |
| 10 | Morphological traces on the parent | Screened beneath the atmospheric gas envelope. |
| 11 | Structural traces on the moons | Preservation of a frozen shape (Fossil bulge) on Iapetus, exceptional structural sponginess on Hyperion, and a density gradient matching the stages of differentiation (detailed below). |
| 12 | Analysis of ring systems | A massive ring system composed almost entirely of pure water ice. Mass measurements made by the Cassini probe indicate that the rings appear young [Iess et al. 2019]; according to the model the ring is a continuous flow fed by an ancient process. The E ring is fed directly by plume activity (geysers) from Enceladus. |
| 13 | Exceptions | The body is not an exception. The exact gradient of density variation among the medium-sized moons is defined for future testing (Part M). |

### Dynamics of the corotation line within the ring system and the ring moons

The main ring system is directly linked to the corotation line ($1.92R$), which passes exactly through the B ring (extending between $92,000\ \mathrm{km}$ and $117,600\ \mathrm{km}$):

- Below the corotation line ($x < 1.92$): the ring material has a negative direction of migration and collapses inward into Saturn. This flow was measured directly and empirically by the Cassini probe as the "ring rain" mechanism (Ring rain) [Waite et al. 2018].
- Above the corotation line ($x > 1.92$): the tiny ring moons – Pan, Atlas, Prometheus, Pandora, Janus and Epimetheus – are located in the range $2.29R - 2.60R$. These bodies have high sponginess and extremely low densities of $400 - 640\ \mathrm{kg/m^3}$ [Porco et al. 2007], representing mechanical remnants of primordial moons that underwent erosion and polishing within the ring, combined with secondary accumulation of ring material on their surfaces.

### The model of the ring as a continuous flow in thermodynamic equilibrium

Since Saturn keeps a spin rate within the birth range and differentiation in the core continues, the hydrodynamic pump is active. The atmospheric cap (70%–85% of the planet's mass)

closes over the water but is not completely sealed, and a little water leaks to the region of the corotation line, where the mass is trapped within the Roche limit and feeds the ring.

Cassini data measured a constant ring-rain rate of hundreds to thousands of kilograms per second [Waite et al. 2018; O'Donoghue et al. 2019]. At this rate of mass removal, without continuous supply, the ring system would break up within a few hundred million years, a finding that led the scientific literature to the assumption that the rings are chronologically young.

According to the uniform mechanism, the ring is not a static accretion remnant but a dynamical system of continuous flow in equilibrium (Steady-state flow): it is continuously renewed by the hydrodynamic pump from below, and removes mass as rain from above. Strict mechanical requirement in the steady state: the rate of radial water leakage from the parent to the ring is exactly equal to the rate of mass removal by ring rain; this mechanism explains why the ring displays the thermodynamic signature of a young body even though the geodynamic process is ancient. Alternative hypotheses of a primordial moon that crossed the Roche limit and broke up mechanically [Wisdom et al. 2022] agree with the model's statement that the ring is a site of destruction and erosion and not a site of construction.

### Chronology of birth and the sequence of differentiation stages on Saturn

All of Saturn's moons formed by a successive ejection mechanism (ejected), Titan, the most massive body, amounting to only $0.024\%$ of Saturn's mass. The order of radial distance reflects the stages of development of the pump in the parent:

| Stage in the parent | Geodynamic mechanism and composition at the opening | Moons formed | Present semi-major axis ($R$) | Calculated density ($\mathrm{kg/m^3}$) |
|---|---|---|---|---|
| **0. Onset of differentiation:** spray of primordial water from the outer layer (wet-rag effect) before the core was sorted. | Iapetus, Hyperion | 61, 26 | 1,088; 544 | |
| **1. Early stage:** ejection of mixed material representing the composition of the inner body as a whole. | Titan | 21 | 1,880 | |

| | | | |
|---|---|---|---|
| **2. Peak of the pump:** supply of the opening with magma rich in water and volatiles from the depths of the sorted body. | Rhea, Dione, Tethys, Enceladus, Mimas | 3 – 9 | 984 – 1,609 |

• **Analysis of stage 0:** the ejection of the water masses took place beyond the local Roche limit of a soft fluid body around Saturn ( $\approx 2.15R$). Iapetus consolidated and froze at an initial distance of $2.53R$, just above this limit, preserving morphological signatures of the birth stage. Its surface is ancient and saturated with craters, which indicates that the body's accretional capture area for surrounding material arriving at a speed of a few hundred metres per second is enlarged by a factor of $1 + v_{\text{escape}}^2/v_{\text{arrival}}^2$ (for Iapetus, where $v_{\text{escape}} \approx 570\ \text{m/s}$, there is an increase of 2 to 5 times or more in the accretion of moving material). The albedo asymmetry of the surface (the dark side) is direct evidence of the landing of orbital dust from the ring of the moon Phoebe onto a body with its own gravitational field [Verbiscer et al. 2009].

• **Analysis of stage 1:** Titan reflects the composition of Saturn's inner body before differentiation was completed. Its moment-of-inertia value ($k = 0.34$ [Iess et al. 2010]) is close to that of Callisto in the Jupiter system (0.355), and sets a lower bound on the density of Saturn's inner body (without the gas envelope), which must be at least as dense as the moon ($1,880\ \text{kg/m}^3$).

• **Analysis of stage 2:** the activity of the pump enriched the outer layer with water, and gave birth to low-density moons made mainly of pure water ice (Tethys, $984\ \text{kg/m}^3$). The thickness of the cap on Saturn allowed prolonged ejection throughout the wet stage, which dictates the system's low mean density compared with the moons of Uranus.

### Geophysical analysis of the stage 0 moons

A. Preservation of the frozen shape of Iapetus

Iapetus is today in full synchronous tidal locking, with an orbital period and spin period of 79 Earth days. However, objective measurements indicate a massive geometric oblateness: the equatorial diameter is 35 km larger than the polar diameter. This deformation matches the hydrostatic equilibrium of a body spinning at an extreme rate of once every 16 hours [Castillo-Rogez et al. 2007], which was preserved owing to rapid thermal morphological freezing.

Since the body was born in a locked state, a spin period of 16 hours requires an orbital period of exactly 16 hours. According to Kepler's laws, this synchronous orbit is obtained at a radius of $147,000\ \text{km}$, which is $2.53R$ from the centre of Saturn (Figure H.1). This finding is direct

geophysical proof that Iapetus consolidated and froze close to its original birth site, just above the primordial corotation line, and was later pushed radially outward to its present position ($61R$). The existence of the massive topographic ridge along the equator (up to 20 km high) is a further frozen structural signature intended for chronological verification.

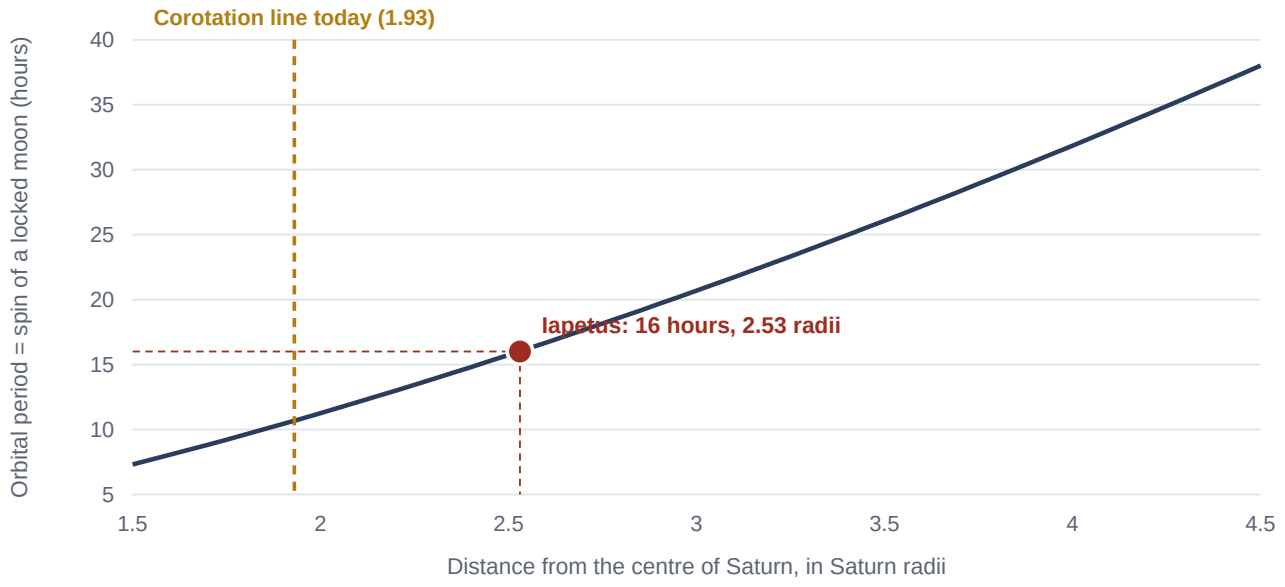



**Figure H.1.** The orbital period around Saturn as a function of distance. A locked moon spins about itself in the same time. The frozen shape of Iapetus matches a spin of 16 hours, that is, an orbit at 2.53 radii, slightly above today's corotation line (the dashed line). *How to read:* find on the vertical axis the spin that the moon's shape remembers, and drop to the horizontal axis to obtain the distance at which the moon hardened.

B. The morphological structure of Hyperion (Hyperion)

Hyperion is an independent body formed at stage 0 (immediately after Iapetus and before Titan) from the water spray of the parent's outer layer, and is not a mechanical fragment of a primordial moon. The body was ejected as a water-laden mass when the corotation line stood at a distance of $x \approx 2.3$ (in the bulge range, as in the Mars system).

According to the cap principle, Saturn's corotation line moved only inward (from $2.5R$ to $1.92R$) owing to the effect of core contraction and the blocking of water escape by the gas. Rapid escape of volatiles and gases into space left the body with extreme structural sponginess (Macro-porosity), giving a mean density of $544\ \mathrm{kg/m^3}$ and no defined core [Thomas et al. 2007], while water-ice and $\mathrm{CO_2}$ components remained frozen on the surface [Cruikshank et al. 2007].

Because of its low mass, Hyperion's self-gravity failed to bring the body to hydrostatic equilibrium, and it keeps an irregular shape. Its orbit is dictated dynamically by the coupling torque of Titan through a strict 4:3 orbital resonance and an eccentricity of $0.1$. The mechanical combination of an irregular shape and an eccentric orbit prevents synchronisation and produces a completely chaotic spin [Wisdom et al. 1984]; the body is located above the corotation line and continues to migrate radially outward.

## PART I Uranus

Uranus is an ice giant with an extreme axial tilt of 98° ("lying on its side"). Its entire system of regular moons moves on circular orbits in the plane of the planet's tilted geometric equator. The system includes five large moons, thirteen small inner moons, and a system of narrow planetary rings.

### Analysis of the 13 protocol checks – Uranus

| # | Protocol check | Findings and numerical values |
|---|---|---|
| 1 | Global basic data | Global mean density: 1,270 kg/m³; present day: 17.2 hours. The internal heat emitted by Uranus is particularly low and close to zero, in marked contrast to the other gas and ice giants [Pearl et al. 1990]. |
| 2 | The system's moon inventory | Five massive moons (Miranda, Ariel, Umbriel, Titania, Oberon [Jacobson 2014]), 13 small inner moons, and 9 captured outer moons (Irregular). |
| 3 | Dynamical reconstruction of the primordial spin | The total orbital momentum of the moons is only about 1% of Uranus's axial spin momentum. Returning this momentum does not significantly change the primordial spin rate, and indicates that the moons themselves did not cause the planet to slow down. |
| 4 | Validation of the birth law | Uranus's corotation line in the present era stands at a distance of 82,700 km from its centre, defined as 3.24R – a value outside the theoretical birth range. For the corotation line to stand within the mechanical range of about 2 radii at the present density, Uranus would have to rotate with a primordial day of 8.3 hours. |
| 5 | Analysis of water and volatile components | The body is extremely rich in water and ice, but lost a massive mass of volatiles through the corotation line owing to the absence of a thick atmospheric cap. |
| 6 | Present degree of differentiation | Partially sorted; a low heat-flow rate indicates geodynamic stagnation. |

| | | |
|---|---|---|
| 7 | Definition of the birth window | The birth window opened with the onset of structural differentiation and closed when the line receded beyond the effective reach of the pump. |
| 8 | Tidal migration path | Integration of the axial tidal forces shows that the tidal mechanism alone can supply the energy to move Ariel and Miranda to their present positions, but fails to move Umbriel, Titania and Oberon. Strict mechanical requirement: a secondary dynamical force pushing outward is required (as for Jupiter). |
| 9 | Dynamical survival parameters | In order within the bounds of the Hill radius. |
| 10 | Morphological traces on the parent | Screened beneath the upper atmospheric envelope. |
| 11 | Structural traces on the moons | Bands around a central block on Miranda, a sign of forced exit from a narrow opening, structural sponginess, and a reversed density gradient (detailed below). |
| 12 | Analysis of ring systems | A system of narrow planetary rings of low mass. |
| 13 | Exceptions | The present position of the line (3.24R) is not an exception but mechanical evidence of massive water escape. The question of the origin of the extreme axial tilt (98°) remains open (Part M). |

**Dynamics of the corotation line leaving the birth range**

The system of the 10 innermost moons (from Cordelia to Perdita) is located today in a radial range of 1.95R - 2.99R – a region lying exactly within the primordial birth range, but below the present corotation line (3.24R). These bodies today have a negative direction of tidal migration and are in inward orbital descent (falling). Their position is direct dynamical evidence that in the past the corotation line stood within the birth range ($x \approx 2$), and receded radially outward over geological history, crossing their orbits.

Since the mass of the moons is tiny (0.01% - 0.02% of the parent's mass) and their orbital momentum does not exceed 1% of Uranus's spin momentum, the slowing of the axial spin from 8.3 hours to 17.2 hours is driven entirely by the escape of water and volatiles through the corotation line:

- Uranus has a thin atmospheric cap (hydrogen and helium make up only 10%–20% of its mass), which failed to trap the water vapour raised by the hydrodynamic pump.

• The water crossed the corotation line and was ejected into space, carrying with it massive angular momentum out of the system ($\Delta L_{\text{escaped}}$).

• **Mechanical lower bound:** to move the line from 2R to 3.24R, a free mass loss of 2%–3% of the mass of the whole planet as water components is required.

• This mass loss is only a lower bound, since since the birth stage a strong inner contraction of the core has taken place in Uranus, which by itself should have spun the planet up and brought the line inward ($x \propto k^{2/3}$, [A.5]); hence the braking torque of the mass of the lost water overcame the effect of structural contraction. The dynamics produce self-regulation: the recession of the line reduces the rate at which water reaches it, and therefore the recession curve is declining (fast at first and slower later).

**Chronology of birth and surface morphology on Uranus**

A. A reversed density gradient compared with the Jupiter system

The density of the five large moons shows an increasing trend as their distance from Uranus increases (the reverse of the trend in the Galilean system of Jupiter):

Miranda (1,200 kg/m³) → Ariel (1,592 kg/m³) → Umbriel (1,390 kg/m³) → Titania (1,711 kg/m³) → Oberon (1,630 kg/m³)

This variation is fully explained by the stages of differentiation of the parent (A.9):

• **Titania and Oberon (born first, on a distant orbit):** ejected at stage 1 (onset of differentiation), and therefore they are dense and reflect exactly the composition of the mixed material of the parent's inner body as a whole.

• **Miranda (born last, on a close orbit):** ejected at stage 2 (peak of the hydrodynamic pump's activity). The body formed as a magmatic mass saturated with free water, which underwent rapid post-birth escape of volatiles that left high structural sponginess and low density (1,200 kg/m³).

• **Inter-system comparison (the two-pot model):** the moons of Uranus are markedly denser than the moons of Saturn (mean silicate mass fraction f = 0.49 versus f = 0.36, [A.9]). The thin atmospheric cap of Uranus allowed free escape of water, which led to the drying and thickening of the parent's upper ejection layer (Figure A.5); therefore, the material scooped from the opening that consolidated into moons was poor in water and dense in mass compared with the saturated material trapped beneath Saturn's thick cap.

B. Morphological signatures of ejection under hydrostatic pressure (grooves and bands)

A planetary body injecting viscous material from a narrow volcanic opening under internal pressure leaves defined mechanical signatures on the surface of the resulting moon:

**1. Straight (linear) directional ejection:** the mass is pushed along a straight line without its own axial spin → formation of an array of parallel grooves (Grooves) along the geometric direction of stretching.

**2. Rotational ejection (mechanical screw mechanism):** the mass is pushed outward while undergoing axial rotation against the walls of the narrow opening → grooves wound around the body, and at the end rings around a centre.

The grooves of Phobos, arranged in parallel planes that cross one another, and the bands of Miranda, which surround a central block (Coronae), match a forced exit from a narrow opening. In the literature the grooves of Phobos are explained as material thrown from craters on Mars [Murray & Heggie 2014], and the bands of Miranda as material that rose from within [Pappalardo et al. 1997].

Both bodies (Phobos and Miranda) have small dimensions and low density, and both represent a cross-section of the peak of the hydrodynamic pump's activity at stage 2 of the parent.

**Vector analysis of the water-escape torque and the axial tilt**

The mechanical analysis rules out the hypothesis that the water-escape mechanism can be the dynamical cause of the extreme axial tilt of Uranus (98°):

- The primordial angular momentum at the moment of birth is defined as a rigid vector (fast axial spin of 8.3 hours, marked as a blue arrow, Figure I.2).
- The present angular momentum is only 48% of the primordial value and is tilted at an angle of 98° (marked as a black arrow).
- For the transition of the system between the two states to be carried out solely by the mass of ejected water, the residual angular-momentum vector of the lost water would have to be 117% of the primordial value, at a tilt angle of 24° relative to the geometric axis (marked as a dashed red arrow).
- **Functional mechanical block:** water ejected through volcanic openings on the geometric equator, and at the height of the corotation line, can carry with it only angular momentum exactly parallel to the planet's axis of spin ($L_{\text{escaped}} \parallel \Omega$). This axisymmetric ejection shortens the length of the global angular-momentum arrow (a vector slowing mechanism) but cannot

rotate or change the direction of the momentum vector in space. Hence, water escape fully explains the slowing of Uranus, but the question of the origin of the extreme axial tilt of 98° remains an open structural question for calculation (Part M).

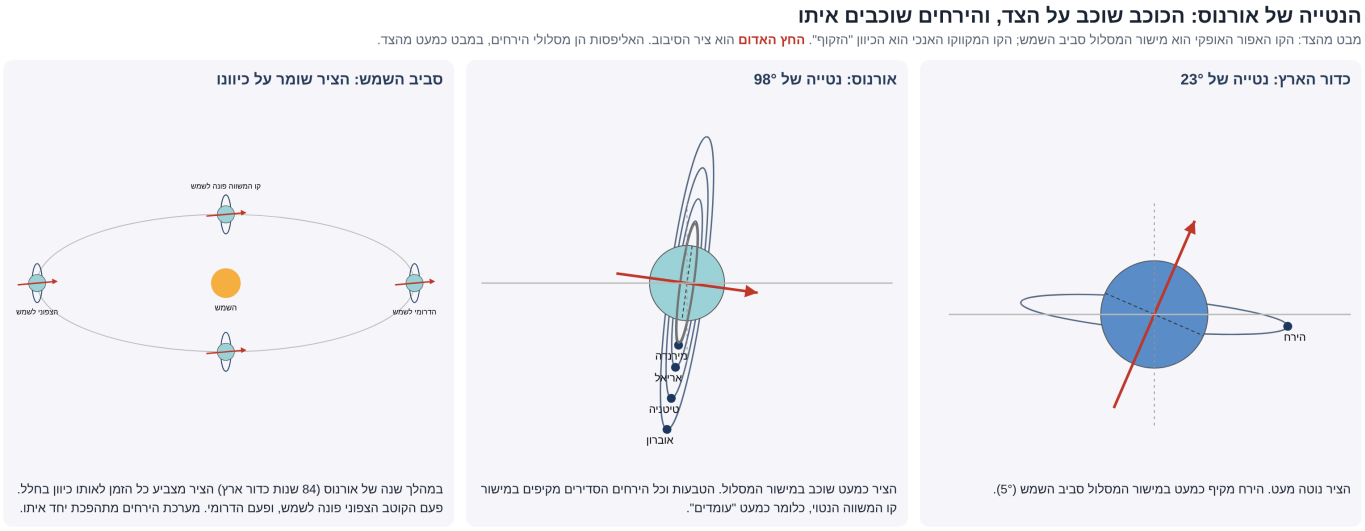



**Figure I.1.** The axial tilt of Earth (23°) and of Uranus (98°), and the position of Uranus's axis along its orbit around the Sun. The regular moons of Uranus orbit in the plane of the tilted equator. *How to read:* Uranus "lies on its side", and therefore along its orbit each pole in turn faces the Sun. The question of the origin of the tilt is open (Part M).

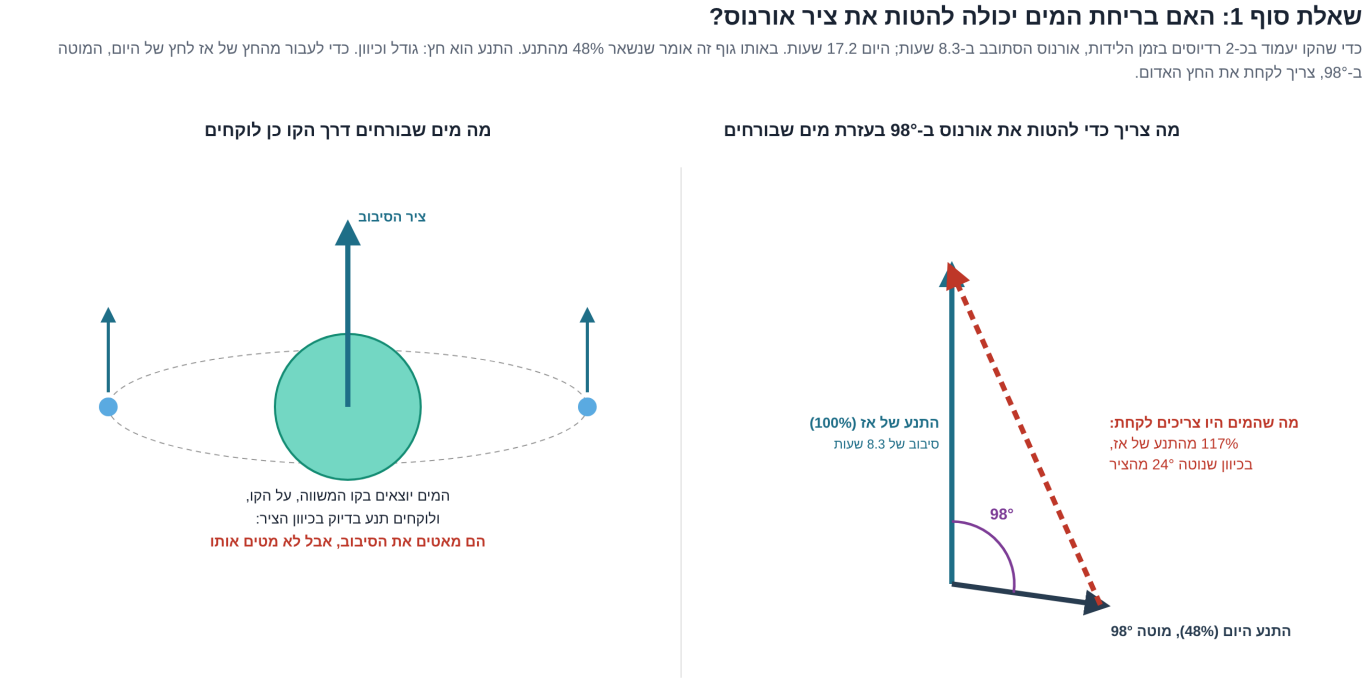



**Figure I.2.** Can water escape tilt Uranus? Right: the momentum then (blue, spin of 8.3 hours), the momentum today (black, 48%, tilted 98°), and what the water would have had to take away to pass between them (dashed red: 117% of the momentum then, in a direction tilted 24° from the axis). Left: water leaving at the equator, at the line,

takes momentum along the axis only. *How to read:* water escape shortens the arrow — the slowing — but does not rotate it.

## I.1 The 98° tilt: possibilities examined

Uranus's spin axis is tilted by 98° from its orbital plane, and all its regular moons orbit in the plane of the tilted equator. This section gathers the dynamical possibilities examined during the research for this paper, the vector calculations carried out, and the competing hypotheses in the scientific literature.

- **The mechanical principle for changing axial tilt:** the angular momentum of a rotating body is defined as a rigid vector in space (having magnitude and direction), and the direction of the axis in space matches the direction of the vector. Changing the tilt angle requires changing the direction of the vector, achieved by adding or removing momentum in a direction not parallel to the axis, or by applying a sustained external torque (Torque). A change in the magnitude of the vector alone (an axisymmetric slowing or spin-up mechanism) changes the spin rate but does not rotate the axis.

**Analysis of the dynamical possibilities and mechanisms**

**Possibility 1: axisymmetric water escape through the corotation line**

At the moment of birth the spin period was about 8.3 hours, and it slowed to 17.2 hours in the present era; Uranus today keeps only 48% of its primordial momentum. A vector transition between the momentum states requires removal of a residual vector of magnitude 117% of the primordial momentum, at a tilt angle of 24° relative to the spin axis (Figure I.2).

- **Mechanical block:** water ejected through the openings of the hydrodynamic pump at the equator carries angular momentum exactly parallel to the axis of spin only. This mechanism shortens the length of the vector (axial slowing) but does not change its spatial direction. The possibility is rejected.

**Possibility 2: one-sided asymmetric mass ejection**

Concentrated ejection of material from a single geographic point located outside the plane of the equator and in a defined direction (like a rocket engine attached to the side of the body) can carry momentum in a direction not parallel to the axis. To supply the required torque (117% of the momentum then), the mass required for ejection was calculated as a function of the ejection velocity:

| Ejection velocity of the material | Mass of material required for ejection | Total mass fraction of Uranus |
|---|---|---|
| Escape velocity (21 km/s) | About 1 Earth mass ($1\ \mathrm{M}_{\oplus}$) | 6.6% |
| Twice the escape velocity (42 km/s) | About 0.5 Earth mass ($0.5\ \mathrm{M}_{\oplus}$) | 3.3% |

• **Mechanical validation:** the orders of magnitude of the mass match the mass of the impactor in the accepted hypotheses ($1 - 3\ \mathrm{M}_{\oplus}$), but act in the reverse direction (momentum carried away by ejection instead of momentum injected by impact). The dynamical limitation lies in the extreme requirement for a concentrated, prolonged ejection mechanism through a single opening and in a fixed direction. The possibility is defined as feasible in terms of the mass budget, but complex in terms of the geodynamic mechanism.

**Possibility 3: displacement of the geometric centre of mass**

When a moon is ejected, the parent undergoes a mechanical recoil in the opposite direction. The displacement of the body's centre from the common centre of mass is defined by the equation: $\Delta r = \frac{m_{\mathrm{moon}}}{M_{\mathrm{Uranus}}} \cdot a_{\mathrm{moon}}$. For Titania the calculation yields Δr ≈ 17 km, for Oberon Δr ≈ 21 km, and for all the moons a few tens of kilometres (negligible compared with a planetary radius of 25,559 km). The spin axis always passes through the centre of mass and is displaced with it, but its spatial direction does not change. An asymmetric distribution of mass forces the body to roll within itself (True Polar Wander), but does not rotate the momentum vector in space. The mechanism alone does not tilt the axis, but it sets a requirement: if the exit of moons is to affect the axis, the moons of Uranus were in the past much heavier than they are today.

**Possibility 4: orbital resonance of the axis precession (Secular Spin-Orbit Resonance)**

Uranus's spin axis undergoes precession (a spinning-top wobble) whose rate is dictated by the flattening moment and the internal mass distribution. Mass orbiting in the plane of the equator behaves, relative to the gravitational field of the Sun, as a direct extension of the planetary bulge. The effective oblateness moment ($J_{2,\,\mathrm{effective}}$) is defined mathematically by:

$$J_{2,\,\mathrm{effective}} = J_2 + \frac{1}{2} \sum_i \frac{m_i}{M} \left( \frac{a_i}{R} \right)^2$$

All the modern moons of Uranus (0.01% of the parent's mass) contribute to the precession rate 5 times the contribution of the planet's own bulge (0.016 versus 0.0033), an effect that holds up to the Laplace radius (53R - 75R). Water ejected by the pump and residing

temporarily in orbit before its removal makes a dominant contribution: a water mass of 2% of Uranus's mass at a radius of 20R increases the effective oblateness moment by a factor of 1,200 relative to the planet's own bulge (Figure I.3).

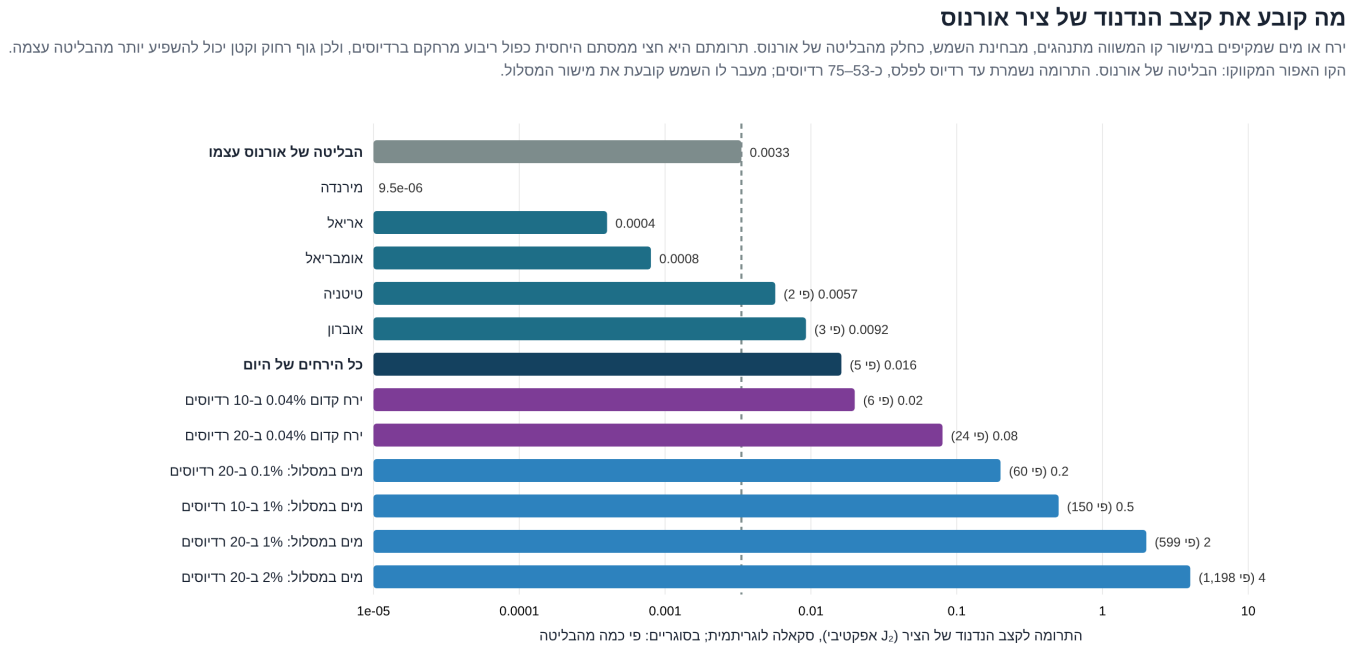



**Figure I.3.** What sets the rate of precession of Uranus's axis. Each column is the contribution to the rate (effective $J_2$) on a logarithmic scale; the dashed line is Uranus's own bulge. Grey: the bulge; dark blue: today's moons; purple: an ancient moon with the minimal mass in the literature; light blue: water lingering in orbit. *How to read:* each column that passes the dashed line affects the precession more than the planet itself.

When the enhanced precession rate meets the external oscillation frequency of Uranus's orbital plane (driven by the neighbouring planets), a strict orbital resonance develops. The Sun's pull on the enhanced bulge tilts the axis gradually, a mechanism parallel to the one that tilted Saturn to 27° [Ward & Hamilton 2004]. The model defines three mass changes that drive this mechanism: the birth of the moons and their recession, the residence of water in orbit, and the contraction of the inner core (A.5). The removal of the water into space breaks the resonance and fixes the final angle. The moons are dragged adiabatically with the plane of the equator and lock their orbits within it. The mechanism fits the model and requires a specific dynamical calculation.

### Possibility 5: a primordial moon that underwent inward tidal collapse

According to the model of Saillenfest and colleagues (2022), the presence of a primordial moon with a mass of at least 0.04% of Uranus's mass (twice the Titan–Saturn mass ratio) migrating outward beyond 10R accelerates the precession frequency until resonance is

reached and tilts Uranus beyond 80°. At this stage the orbit loses dynamical stability, and the moon collapses and falls inward into the parent, which causes the tilt angle to freeze immediately.

Within the uniform mechanism, this hypothesis is a particular case of possibility 4, and provides a physical explanation for the fate of the body: the moon fell back after the receding corotation line crossed its orbit, like the dynamics of the 10 inner moons today. Analysis of the falling stage shows that the return of the moon to the equator of a soft parent causes a displacement of the mass distribution and a shift of the principal flattening moment (C-A):

| **Mass of the collapsing moon ($m_{\mathrm{moon}}/M_{\mathrm{Uranus}}$)** | **Mass fraction displaced from the principal flattening moment (C-A)** |
|---|---|
| Minimal mass in the literature (0.04%) | About 12% |
| Preferred mass in the literature (0.17%) | About 51% |

This mass displacement changes the precession frequency abruptly and breaks the resonance. If the fall took place from an inclined orbit, a secondary angular-momentum component is injected in a direction not parallel to the axis. A matching mechanism that requires calculation of the dynamical timescales and of the strength of the morphological amplification.

### Analysis matrix of the competing hypotheses and models

| # | The dynamical hypothesis | Mechanical requirements and strict parameters | Fit with the uniform mechanism | Functional status |
|---|---|---|---|---|
| 1 | Water escape through the corotation line | Carrying 117% of the angular momentum in a direction not parallel to the spin axis. | — | Rejected. The ejection torque is parallel to the axis only. |
| 2 | One-sided asymmetric mass ejection | Ejection of $0.5-1\ \mathrm{M}_{\oplus}$ from a single geographic point and in a fixed direction. | Yes (driven by the hydrodynamic pump). | Quantitatively possible; complex in terms of the mechanism focusing the opening. |
| 3 | Displacement of the centre of mass alone | Displacement of the core geometry from the common centre of mass. | — | Rejected. Does not change the direction of the vector in space. |
| 4 | Resonance from changes in mass | The effective oblateness moment reaching a | Yes, fully. (Combination of | Fits the mechanism; requires a dynamical |

| | | | | |
|---|---|---|---|---|
| | distribution | precession frequency matching the external frequency of the system. | moons, water in orbit and contraction.) | calculation of the time windows. |
| 5 | A massive primordial moon that collapsed | Moon mass ≥ 0.04%, radial migration > 10R and post-resonance inward collapse. | Yes, fully. (Matches the mechanism of line crossing and sponginess.) | Fits the mechanism; requires calculation of the geodynamic amplification factor. |
| 6 | Giant impacts (the accepted explanation) | Direct mechanical impact of a body of size $1-3\ \mathrm{M}_{\oplus}$; requires several successive impacts [Morbidelli et al. 2012]. | No. A competing mechanical model. | Accepted in the literature; suffers from a compound probability problem. |
| 7 | Resonance during the planetary migration stage | Frequency coupling with a neighbouring planet during the early evolutionary stage [Boué & Laskar 2010]. | Neutral. | Limited; theoretical failure to reach the extreme angle of 98° without a secondary force. |

**Summary and empirical tests for verification**

The uniform model does not decide finally the question of the origin of the axial tilt, but it establishes two clear geophysical conclusions: the water-escape mechanism (which explains the slowing of the axial spin and the recession of the corotation line) cannot tilt the direction of the axis; by contrast, structural changes in mass distribution (the presence of water and moons on distant orbits in the plane of the equator) control the precession frequency with a strength much greater than that of the planet's inner body, and are therefore the most natural physical mechanism within the model.

**The set of tests and predictions for verifying the model:**

**1. Mapping of the gravity field (deep gravimetry):** a massive primordial moon that underwent mechanical collapse into the interior of a soft parent (possibility 5) left a defined asymmetric gravity anomaly – an excess of mass at the equator on one side.

**2. Calculation of the historical precession rate:** running the differential equation of motion along the chronology of births and water escape, for mathematical validation of the crossing of the resonance window and of the residence time required to reach an angle of 98°.

Uranus preserves the pattern of both sides of the line and the morphological signature of tidal locking. The inner moons falling today validate geometrically the position of the corotation line

at the moment of birth, and the water escape establishes its radial recession, while the question of the mechanism of the 98° axial tilt is defined as an open problem for further calculation (Part M).

## PART J Neptune

Neptune is the most distant and the densest ice giant in the solar system. The system has seven small inner moons and a single massive moon, Triton, which has a reverse orbit (Retrograde). The other outer moons are defined as captured moons (Irregular) and are not included in the protocol checks.

### Analysis of the 13 protocol checks – Neptune

| # | Protocol check | Findings and numerical values |
|---|---|---|
| 1 | Global basic data | Global mean density: 1,638 kg/m³; present day: 16.1 hours. The spin axis is tilted at an angle of 28°. |
| 2 | The system's moon inventory | Seven small inner moons, a single massive moon (Triton), and a group of captured outer moons. |
| 3 | Dynamical reconstruction of the primordial spin | The orbital momentum of the inner moons is negligible. The post-birth migration torque is entirely dominated by the system's guest, Triton, which exerts constant braking. |
| 4 | Validation of the birth law | Neptune's corotation line in the present era stands at a distance of 83,500 km from its centre, defined as 3.39R – outside the theoretical birth range (as in the Uranus system). For the present density, the birth law requires a primordial day of 7.3 hours. |
| 5 | Analysis of water and volatile components | The body is rich in ice and water components; it underwent a massive loss of momentum and of water mass through the corotation line owing to a thin atmospheric cap. |
| 6 | Present degree of differentiation | Sorted. |
| 7 | Definition of the birth window | Closed with the displacement of the corotation line beyond the active range of the hydrodynamic pump. |
| 8 | Tidal migration path | Dictated dynamically by the unique orbit of Triton. |
| 9 | Dynamical survival parameters | The inner system is stable; Triton has a constant negative direction of migration (falling inward because of its reverse orbit). |

| | | |
|---|---|---|
| 10 | Morphological traces on the parent | Screened beneath the deep atmospheric layers. |
| 11 | Structural traces on the moons | Preservation of an irregular shape in bodies on the verge of rounding (Proteus) and a high density in Triton (detailed below). |
| 12 | Analysis of ring systems | A system of tenuous rings and asymmetric ring arcs (Ring arcs). |
| 13 | Exceptions | The moon Triton is a morphological and dynamical exception to the regular system, fully explained as a body captured from outside. |

**Dynamics of the corotation line and the morphology of the moon Proteus**

The system of inner moons preserves exactly the pattern of both sides of the line of the uniform model (Part L):

- **Below the present corotation line (x < 3.39):** the five moons Naiad, Thalassa, Despina, Galatea and Larissa are located in a radial range of 1.96R - 2.99R; they lie below the line, have a negative direction of migration and are in inward orbital descent (falling back).
- **Above the present corotation line (x > 3.39):** the moons Hippocamp [Showalter et al. 2019] and Proteus are located above the line.

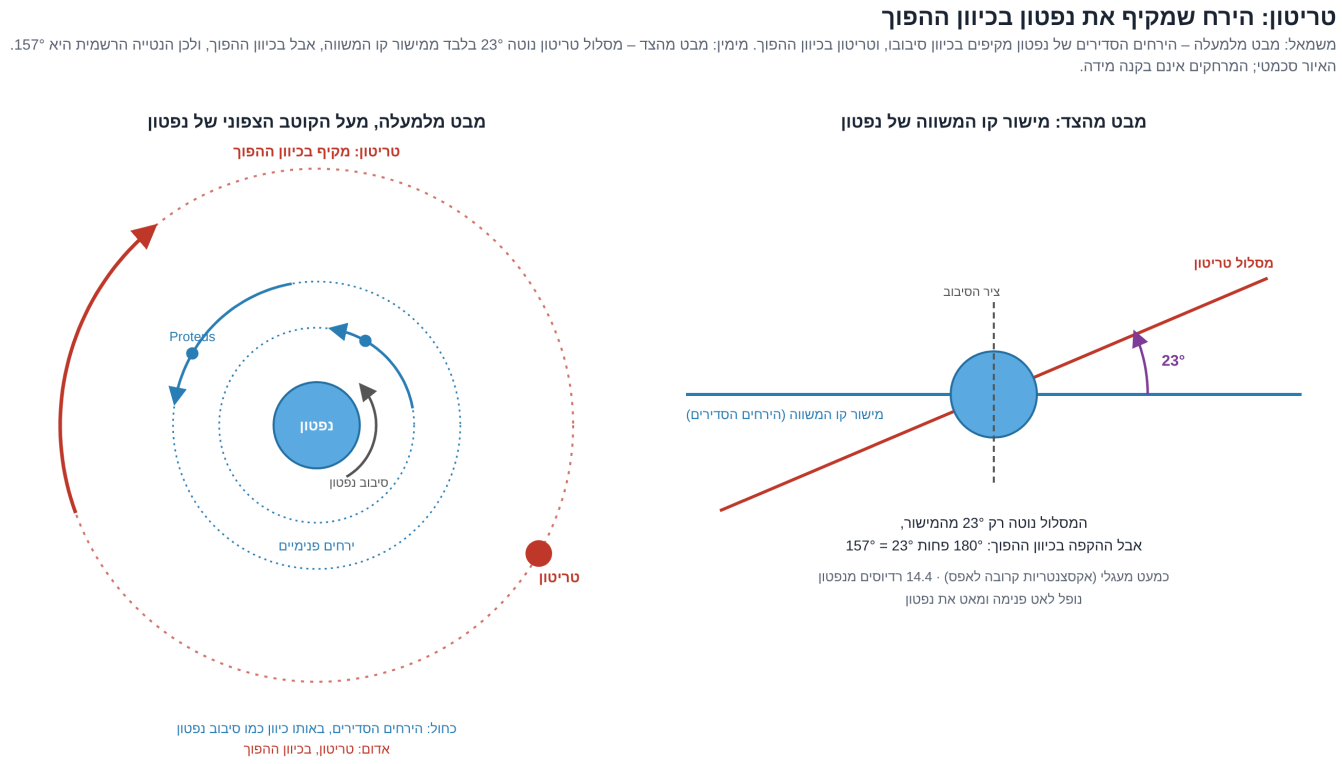
The slowing of Neptune's axial spin from a primordial day of 7.3 hours to 16.1 hours is explained, as for Uranus, by the mechanism of escape of water and volatiles through the corotation line ($\Delta L_{\text{escaped}}$), which carried essential angular momentum out of the system and moved the line outward. The inner moons residing today within the primordial birth range and falling inward are dynamical proof that the line crossed their orbits during its recession.

- **The structural parameters of Proteus:** the moon has a mean diameter of about 420 km, a size lying exactly at the theoretical rounding threshold of an icy body (about 400 km, [A.4]). Nevertheless, the body keeps an irregular asymmetric shape, has an extremely dark surface, and carries a massive impact crater (Pharos). According to the uniform mechanism, the body is a spongy body (Macro-porosity) ejected as a water-saturated magmatic mass; the rapid escape of volatiles caused rapid thermal hardening of the silicate-icy framework before the body could reach hydrostatic equilibrium and become round. Geophysical prediction: the density of Proteus (not yet measured empirically) must be exceptionally low, as evidence of the internal voids of stage 2.

**Triton: mechanical analysis of a body captured from the Kuiper belt**

## A. The dynamical and chemical parameters

Triton orbits Neptune on a reverse orbit (Retrograde) at a constant orbital velocity, with an extreme orbital inclination of 157° (equivalent to an angle of only 23° from the plane of the planetary equator, but in the reverse direction, Figure J.1). The orbit is perfectly circular at a radius of 14.4R.




| תכונה | מה צפוי מירח שנולד על קו הקורוטציה | טריטון | מתאים? |
|---|---|---|---|
| כיוון ההקפה | כמו סיבוב ההורה | הפוך | ✗ |
| נטייה ממישור קו המשווה | כמעט אפס | 157° (23° במהופך) | ✗ |
| צורת המסלול | כמעט מעגלית | מעגלית כמעט לחלוטין | ✓ |
| מרחק | טווח הלידה, או מקום שאפשר לנדוד אליו | 14.4 רדיוסים | ? |
| הרכב וצפיפות | כמו האחים וחומר ההורה | קרח חנקן; 2,061 – כמו פלוטו (1,854) | ✗ |
| גודל ביחס לאחים | ירחי הענקים קטנים | כ-99.5% ממסת כל ירחי נפטון | ? |

ירוק: מתאים; אדום: אינו מתאים; צהוב: לא מכריע. לפי הכלל המוצע ("לפחות שניים משלושה סימנים: כיוון או נטייה, מרחק, הרכב") טריטון מסומן כנלכד בשני סימנים – כיוון והרכב. ההכרעה במקומו במנגנון פתוחה (חלק M).

**Figure J.1.** Triton. Left: view from above Neptune's north pole — the regular moons orbit in the direction of rotation, and Triton in the reverse direction. Right: side view — the orbit is inclined only 23° from the plane of the equator, but the orbit is reversed (157°). Bottom: the properties of Triton compared with what is expected of a moon born on the corotation line. The figure is schematic.

The body has a physical radius of 1,353 km and a mass of $2.14 \times 10^{22}$ kg – a value amounting to 0.021% of Neptune's mass (a mass ratio on the scale of Titan–Saturn, 0.024%). Its dimensions and mass are greater than those of Pluto. The surface is composed of nitrogen, methane and carbon monoxide ice, and shows cryovolcanic activity (nitrogen geysers, [Smith et al. 1989]). Its mean density is high, at 2,061 kg/m³.

Since the birth law requires that a moon released from the parent move in the direction of its axial spin and in the plane of its geometric equator, Triton is not an original moon of Neptune.

B. Reconstruction of the body's origin (the model of a lobe of a peanut that split)

Comparative analysis of the body's mass and density rules out the possibility that Triton is a moon pushed from the Uranus system (heavier than all the moons of Uranus together, 6 times Titania) or from the Pluto system (heavier than Pluto). The body fits exactly as a massive member of a binary body system from the Kuiper belt, which was captured dynamically by Neptune's gravitational field while its partner was cut off and thrown into space [Agnor & Hamilton 2006].

According to the uniform mechanism, this binary system formed at the outer edge of the solar system through the third mode of birth – a full hydrodynamic split (Fission) [A.4, K.1]:

- **The primordial united body underwent intense differentiation (A.5):** the dense silicate and metallic material sank to the system's centre of mass, located structurally within the large lobe of the two-lobed peanut structure.
- As the spin rate rose beyond the break-up threshold, the geometric neck was cut through outside the global centre of mass.
- As a result, the large lobe (Triton) captured and kept most of the dense silicate and metallic core, while the small lobe left as a body rich in ice and volatiles and of low density. Triton represents the large lobe of a primordial peanut system that was more massive than Pluto–Charon and underwent more advanced differentiation, which makes it denser and more massive than Pluto.

The capture of Triton caused disruption and dynamical collapse of Neptune's original moon system [Goldreich et al. 1989], a mechanism that explains the complete absence of medium-sized regular moons in the present system.

**Analysis of peanut systems that split in the Kuiper belt**

The geodynamic model defines a strict structural prediction for all binary bodies formed by the split mechanism: the massive member of the system (the parent) is necessarily denser than the secondary member (the moon), owing to the concentration of the core in the large lobe before the neck was torn.

| Primordial binary system | Mass ratio ($m_{\text{small}}/M_{\text{large}}$) | Density of the large body ($\text{kg/m}^3$) | Density of the small body ($\text{kg/m}^3$) | Validation of the geophysical prediction |
|---|---|---|---|---|
| Pluto / Charon | 0.12 | 1,854 | 1,702 | Fully confirmed. The large body is denser than the small one. |
| Eris / Dysnomia | < 0.0085 | 2,400 | $700 \pm 500$ | The moon is of low density, and fits better a body ejected by the spin than a lobe [Brown & Butler 2023]. |
| Orcus / Vanth | 0.16 | $1,400 \pm 200$ | 1,500 (range: 1000-2500) | Not yet decided. Requires an exact measurement of the density of Vanth. |

Competing hypotheses in the scientific literature interpret the formation of these binary systems as the product of primitive giant impacts. The density gradient in which the massive body is denser than the secondary is also geometrically consistent with the impact model, but the decisive mechanical evidence in favour of the hydrodynamic split mechanism lies in the fact that the total angular momentum of the primordial united body of Pluto–Charon stands beyond the mechanical break-up threshold (K.1).

## PART K Dwarf planets (?)

The planetary bodies in this part are marked with a methodological question mark, since dynamically some of them may be primordial moons that underwent orbital escape or gravitational capture. The dynamical system of Triton proves that an independent body from the Kuiper belt can become a moon; on the other hand, some of these bodies have their own moon systems, and they act mechanically as parents.

### Global and structural parameters of the dwarf planets

| Planetary body | Mean density ($\text{kg/m}^3$) | Spin period (day) | Present position of the corotation line ($R$) | The system's moon inventory | Moon/parent mass ratio |
|---|---|---|---|---|---|
| Pluto | 1,854 | 6.4 days (synchronously locked) | 16.5R | Charon + 4 small moons | 12% |
| Haumea (Haumea) | 1,882 | 3.9 hours | 1.38R | 2 moons + ring system | 0.45% |

| | | | | | |
|---|---|---|---|---|---|
| Eris (Eris) | 2,519 | 15.8 days (synchronously locked) | 32R | Dysnomia (Dysnomia) | 0.5% |
| Makemake | 2,025 | 22.8 hours | 4.6R | A single tiny moon | 0.03% |
| Ceres (Ceres) | 2,157 | 9.1 hours | 2.54R | No moons | — |

## K.1 Pluto and Charon: the peanut that split

Charon amounts to 12% of Pluto's mass [Brozović et al. 2015], the highest mass ratio in the solar system for a planetary moon (fraction of the system's mass: Pluto 89%, Charon 11%). According to the uniform mechanism, Pluto and Charon were a single united body – a two-lobed peanut structure (Contact binary) – which underwent a full hydrodynamic split (Rotational fission) at its geometric neck. Two-lobed bodies of this kind are structurally common in the Kuiper belt, as seen in the geometric structure of the body Arrokoth [Stern et al. 2019].

A. Vector analysis of the spin and split thresholds

Full computational return of the system's angular momentum into a primordial united body (calculated radius: about 1,240 km, density: about 1,840 kg/m³) defines an axial spin period that depends on the degree of structural differentiation. For a body of this density, two strict thresholds are set by the Maclaurin and Jacobi equations [Chandrasekhar 1969]:

1. The ellipsoidal elongation threshold ($P \leq 4.6$ hours): the body cannot keep a symmetric oblate shape, and is stretched into a triaxial ellipsoid and later into a peanut structure.

2. The mechanical separation threshold ($P \leq 2.44$ hours): the equatorial velocity equals orbital velocity ($x = 1$), and the body is torn apart.

### Effect of differentiation on reaching the split threshold

| State of the primordial united body | Moment-of-inertia factor ($k$) | Calculated spin period ($P$) | Position of the corotation line ($x$) | Status relative to the mechanical thresholds |
|---|---|---|---|---|
| Uniform homogeneous | 0.4 | 2.51 hours | 1.02 | Beyond the elongation threshold, close to the |

| | | | | |
|---|---|---|---|---|
| (before differentiation) | | | | mechanical separation threshold. |
| Structurally sorted (active differentiation) | 0.31 | 1.94 hours | 0.86 | Full crossing of the separation threshold (tearing). |

The united body had an elongated morphology even before sorting began. It was the process of structural differentiation that drove the full split: the sinking of the dense silicate materials into the core reduced the moment-of-inertia factor to 0.31 and accelerated the axial spin under conservation of momentum [A.5]. The corotation line stood right at the surface, at the height of the geometric neck, and caused the structure to tear.

Supporting empirical findings: Pluto and Charon are in full mutual tidal locking (Double synchronous lock) [Stern et al. 2015], and Charon is located exactly on Pluto's corotation line, evidence of two bodies that split on the dynamical line. Their densities are close (1,854 versus 1,702 kg/m³), a signature typical of homogeneous material that split at the early stages of differentiation (stage 1, [A.9]).

B. Mechanics of the morphological fracture on Charon

Pluto is located in the frozen zone of the planetary arc of births [L.4], a structure of a soft subsurface core blocked beneath a static frozen crust. The crust showed elastic resistance to the rotational tensile stresses; as inner contraction advanced and the spin rate rose, the outer crust cracked at the neck all at once by a mechanical fracture mechanism (Brittle failure). The hydrostatic pressure in Pluto's core (about 0.7 GPa) is 700 times higher than the maximum tensile strength of ice, and therefore the fracture was limited to the outer crustal layer and released the tensile stresses at the mechanical threshold:

- **The Oz Terra region (northern hemisphere):** has a rough, ancient topography (geological age > 4 billion years), which carries the signature of the original crust of the peanut stage.
- **The Vulcan Planitia plain (equatorial region):** a completely smooth plain representing the soft inner material exposed at the moment of fracture. The mass flowed and rounded back to hydrostatic equilibrium, a stretching process that smoothed the surface. The isolated mountain structures lying in depressions in the plain are blocks trapped within the soft matrix during the stretching. The plain is located exactly on the equator, in the hemisphere permanently facing Pluto (longitude orientation 285° to 30° [Beyer et al. 2017]), the site of the primordial neck.

• **The tectonic canyon belt:** the line of ridges and rifts separating the two geological units is, according to the model, the mechanical fracture line of the split. In the literature the belt appears to encircle the whole of Charon, and is explained as extension while an internal ocean froze [Beyer et al. 2021].

C. The orbital dynamics of the four small moons

The four small moons – Styx, Nix, Kerberos and Hydra – were ejected directly from the equatorial and neck region of the primordial peanut before the final split stage, when the spin rate reached the separation threshold and the corotation line was locked at the surface. In this dynamical configuration, the equator loses mass by itself owing to centrifugal forces, without the need for a point volcanic pump. The moons were released on circular orbits in the plane of the geometric equator.

The empirical measurements of the New Horizons mission validate this mechanism [Weaver et al. 2016]:

• The orbits are perfectly circular and in the plane of Charon's synchronous orbit.

• The bodies are tiny and not in hydrostatic equilibrium (asymmetric).

• The surface albedo is extremely high, made of clean water ice, a signature of wet inner material ejected into space.

• The orbital periods show orbital resonance ratios (Resonance) close to 1:3, 1:4, 1:5 and 1:6 relative to Charon's orbital period, with Styx, Nix and Hydra locked in a strict three-body resonance (Laplace-like resonance) [Showalter & Hamilton 2015].

D. The two-stage model for the migration of the small moons

From the moment of their detachment, the small moons were placed above the dynamical corotation line and had a positive direction of migration (they receded radially outward). The recession mechanism operated in two successive stages:

Stage A: push by the primordial peanut (quadrupole moment) → Stage B: global tidal push (soft Pluto body)

• **Stage A (push by the peanut):** an elongated two-lobed body spinning at high speed produces a massive asymmetric gravitational potential field (a strong quadrupole moment), which exerts an outward pushing force several orders of magnitude greater than classical tidal forces at the birth sites.

• **Stage B (global tidal push):** as the split advanced, the push was converted into the tidal forces of young Pluto. The strength of the push declines sharply as a function of distance ($\dot{a} \propto a^{-11/2}$); doubling the distance reduces the displacing force by a factor of about 45. Close to the surface, for a body of high mechanical softness ($k_2/Q \approx 0.1$), the migration rate was hundreds of metres per year, and declined to zero in the modern era.

Time integration of stage B alone (assuming a soft Pluto throughout the whole geological period) moves Nix and Hydra to a radial distance of 17R - 23R within 4.5 billion years. This position passes the stopping point of the present corotation line ($16.5R$, fixed by the locking of Pluto's spin to Charon's orbit), and it is thus proven dynamically that the corotation line never caught up with the small moons. Since the bodies are today in the range 36R - 55R, a strict mechanical requirement is set: the quadrupole pushing torque of the peanut structure in stage A closed a gap of a factor of 2 in orbital distance. Hypotheses of orbital dragging by Charon [Ward & Canup 2006] are rejected, since this mechanism increases orbital eccentricity, contrary to the circular orbits of the present era [Cheng et al. 2014].

## K.2 Haumea (Haumea): a peanut before splitting

The dwarf planet Haumea has an extreme spin period of 3.9 hours [Rabinowitz et al. 2006] and an exceptionally elongated ellipsoidal morphology ($2,300 \times 1,100$ km [Ortiz et al. 2017]). Its corotation line is located at a radius of 1,104 km, while the tips of its long geometric axis extend to a distance of 1,161 km; the tips of the physical body extend beyond the present corotation line. The body has a planetary ring system at 2,287 km [Ortiz et al. 2017], two regular moons, and a dynamical family of icy bodies with similar orbital elements [Brown et al. 2007].

According to the uniform mechanism, Haumea is the body in the modern solar system closest to a stage of active physical birth in the present time:

• For its density ($1,882$ kg/m$^3$), the ellipsoidal elongation threshold is at about 4.5 hours, and the mechanical separation threshold at about 2.4 hours.

• The present spin period (3.9 hours) lies exactly within this window.

• The body represents a pre-split peanut structure, a cross-section of the geological stage that Pluto and Charon passed through before the neck tore. Geodynamic prediction: continued contraction of Haumea's inner core will bring about axial spin-up by conservation of momentum, and will push the body toward the mechanical separation threshold at 2.4 hours [A.5].

### K.3 Eris and Dysnomia (Eris & Dysnomia)

The Eris system is in full synchronous tidal locking with the moon Dysnomia [Szakáts et al. 2023], which is located exactly on the parent's static corotation line ($x = 32$). The mass ratio of the system is about 0.5% [Brown & Butler 2023]. The dynamical and structural configuration reproduces exactly the Pluto–Charon model on a smaller mass scale, and matches the statements of the moon law for the advanced stages of the arc of births at the outer edge.

### K.4 Makemake

Makemake has one small moon (Parker et al. 2016). Its corotation line today is at 4.6 radii. The data are scarce.

### K.5 Ceres (Ceres)

The dwarf planet Ceres is a body rich in water components that underwent only partial structural differentiation during its geological history [Park et al. 2016]. The corotation line of Ceres in the present era stands at a distance of $2.54R$ from its centre, a value located markedly close to the theoretical birth range of the uniform model.

Despite the parallel physical presence of the water conditions and the axial spin parameters, the body is completely moonless. This mechanical deviation from the moon law is not decided at this stage, and is defined as an open structural question for further research (Part M).

### K.6 Dwarf or moon?

The dynamical analysis of the outer edge of the solar system points to a duality of definition (Dual nature) for the celestial bodies in the dwarf-planet category:

- **Evolutionary path A:** an independent body becomes a moon through gravitational capture (Capture). This mechanism is established empirically in the Neptune system, where the geophysical and orbital data define Triton as a dwarf planet from the Kuiper belt captured from outside.
- **Evolutionary path B:** an independent body acts as a parent and gives birth to a moon system of its own. Pluto and Eris (Eris) represent this mechanism, and behave physically as parents in every respect.

The theoretical concern that some of the dwarf planets recognised today are no more than primordial moons that underwent orbital escape (Escaped moons) or secondary capture is

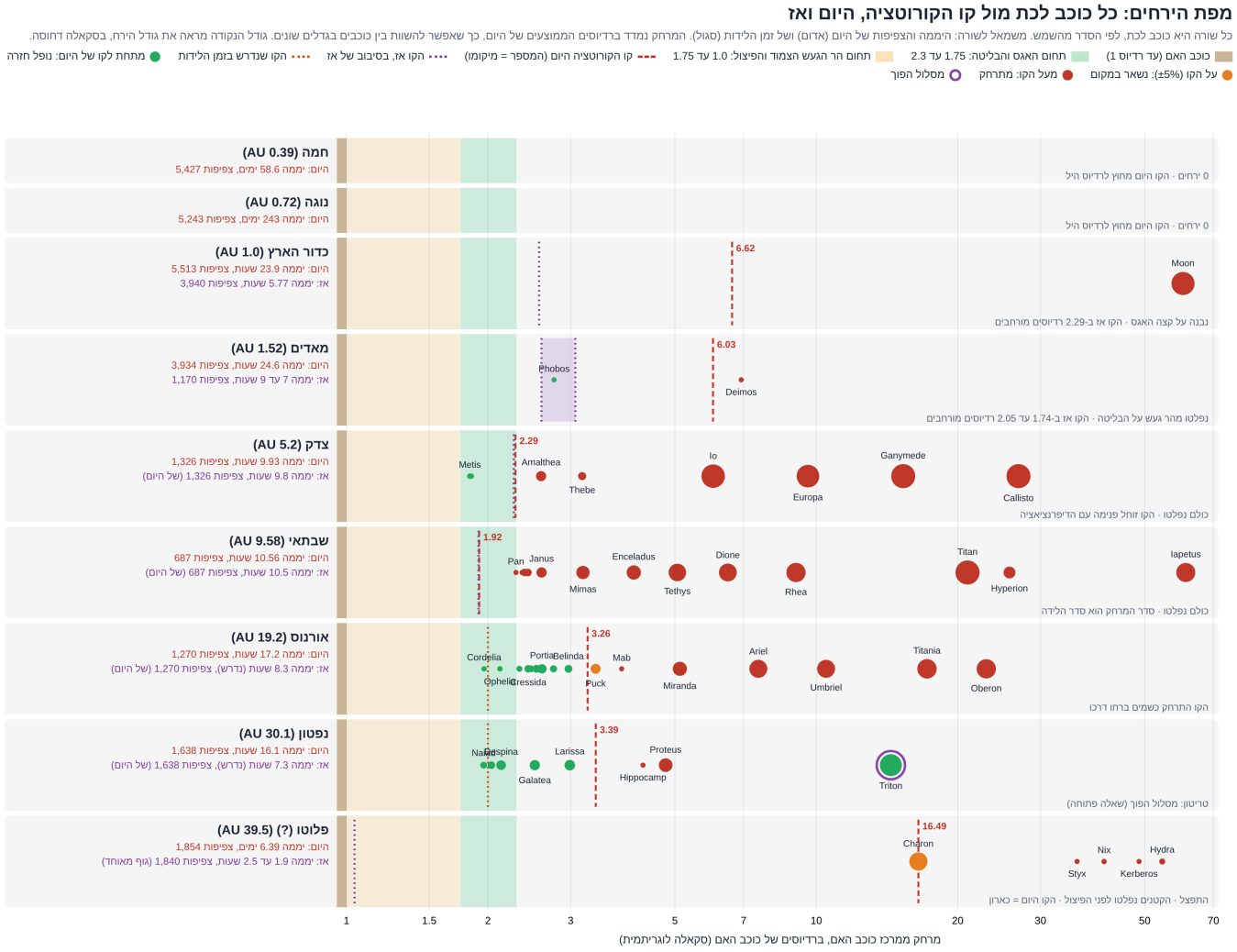
mechanically justified. Establishing absolute geodynamic criteria for distinguishing between an original dwarf planet and a primordial moon that escaped is defined as an open structural problem (Part M).

## PART L The common denominator

The unifying part brings all the celestial bodies examined in the solar system together under one parametric matrix, in order to validate the uniform mechanism and present the recurring dynamical signatures.

### L.1 Moons on both sides of the line

In five planetary systems of the solar system – Mars, Jupiter, Saturn, Uranus and Neptune – a uniform dynamical anomaly is identified: a group of small regular moons located in the immediate vicinity of the parent's present corotation line, distributed on both sides of it (Figure L.1):




בכדור הארץ ובמאדים הכוכב היה מורחב ברגע הלידה (7,157 ק"מ ו-5,085 ק"מ), ולכן ברדיוסים של היום הקו של אז נראה רחוק יותר מטווח הלידה; ברדיוסים של אז הוא יושב בתוכו. בצדק ובשבתאי הקו של אז מחושב מהחזרת התנע הזוויתי של הירחים, והצפיפות היא של היום. באורנוס ובנפטון: הסיבוב שנדרש כדי שהקו יעמוד בכ-2 רדיוסים. בפלוטו: פלוטו וכארון כגוף אחד (רדיוס כ-1,240 ק"מ), והקו כמעט על פני השטח. מחוץ לתחום האיור: Phoebe של שבתאי ו-Nereid של נפטון. באורנוס הושמטו Cupid, Perdita ו-Belinda; בשבתאי הושמטו Epimetheus, הצמוד ל-Janus.

**Figure L.1.** Map of the moons relative to the corotation line. Each row is a planet, from the closest to the Sun (top) to the most distant (bottom). Distance is measured in parent radii, on a logarithmic scale. The green band is the birth range; the dashed red line is the corotation line today; the dotted purple line is the line then, at the spin of that time; the dotted orange line for Uranus and Neptune is the line required at the time of the births. Green moons are below today's line and are falling back; orange ones sit on the line; red ones are above it and receding. Triton is marked with a purple ring (reverse orbit). *How to read:* in each row, look for the red line; in all systems there are moons on both sides of it, and for Uranus and Neptune the green moons sit within the birth range, where the line stood at the time of the births.

- **The sub-synchronous group (below the line):** has a negative direction of tidal migration and creeps in inward orbital retreat toward collapse onto the parent (green moons). In the Uranus and Neptune systems, these moons sit exactly within the primordial birth range ($1.95R - 3.0R$).
- **The super-synchronous group (above the line):** has a positive direction of tidal migration and recedes radially outward (red moons).

This symmetric configuration is direct mechanical proof of the prediction of the moon law: the bodies were born synchronously on the corotation line, and underwent two-way dynamical dispersal according to the direction of tidal migration on the side on which they consolidated. On Uranus and Neptune, the sub-synchronous moons falling today fix the original geometric position of the corotation line at the moment of birth, before it was moved outward by the escape of water.

## L.2 The common-denominator matrix

The research matrix (Figure L.2) crosses 13 celestial bodies (planets and dwarf planets) with nine physical criteria of the uniform mechanism, and summarises the degree of empirical fit of the model to present reality.

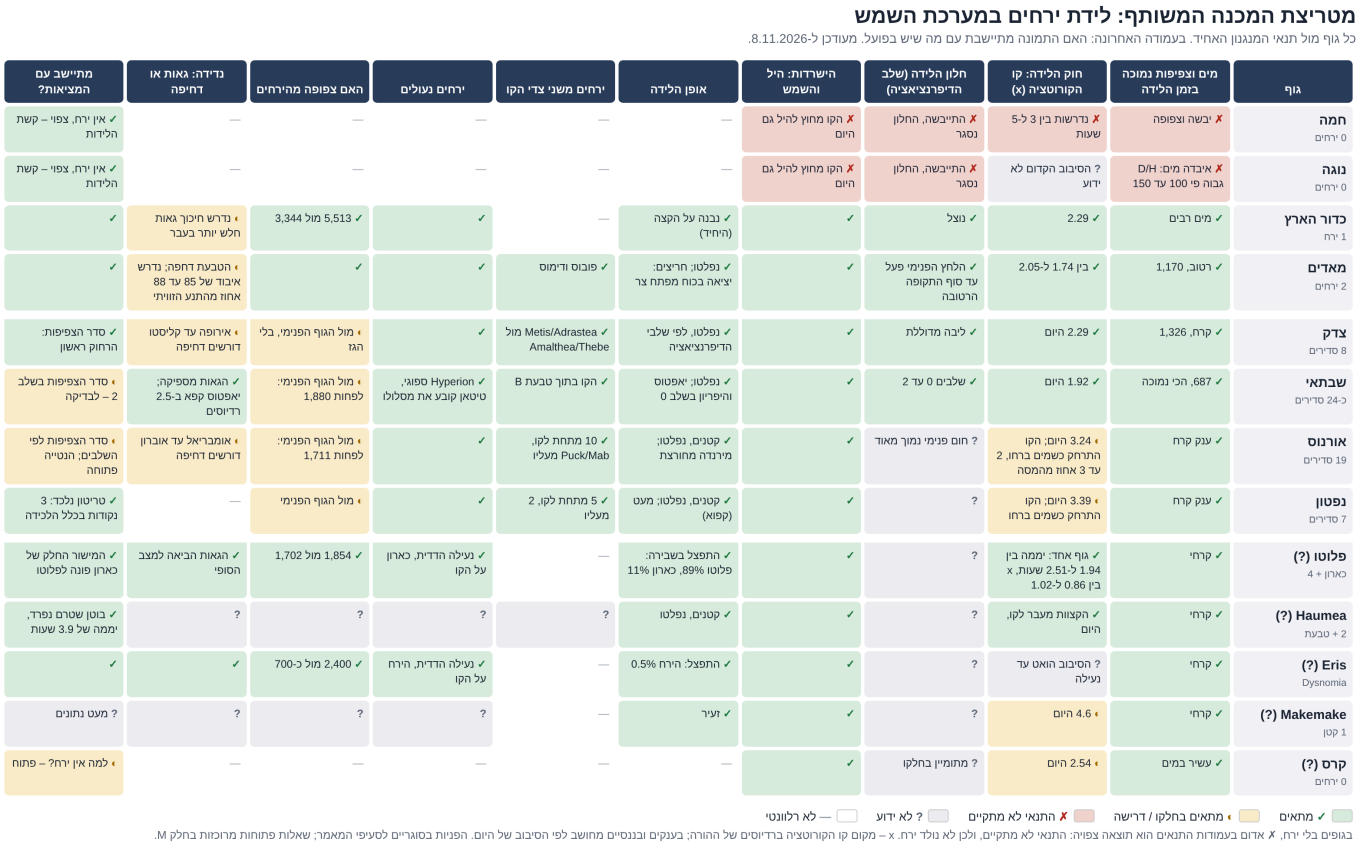



**Figure L.2.** The common-denominator matrix: 13 bodies against nine conditions, updated for all the decisions in the paper, and in the last column — whether the picture agrees with what actually exists. Green: fits; yellow: partly fits or a requirement; red: the condition is not met; grey: unknown. For bodies without a moon, red in the condition columns is an expected result: the condition is not met, and there is no moon.

## Indicators of the system matrix:

- **Green status (full fit):** the measured parameters of the body match exactly the predictions of the moon law.
- **Yellow status (partial fit / mechanical requirement):** the body fits the model provided a strict dynamical requirement is met (such as loss of angular momentum on Mars and Uranus, or an external push of moons on Jupiter and Uranus), without physical contradiction.
- **Red status (the condition is not met):** applied to moonless bodies (Mercury, Venus), in which the absence of the criterion directly explains why birth did not occur, as expected from the arc of births.
- **Grey status (unknown parameter):** missing data not yet measured empirically (such as the density of the moon Proteus at Neptune).

## Summary of the matrix findings and the radial evolution

The global analysis proves that the uniform mechanism is valid and continuous throughout the solar system, with no bodies that contradict the basic laws. The morphological variation between the systems reflects a continuous function of radial distance from the Sun, which defines the planetary arc of births (L.4) and the division into the three modes of birth:

Inner edge: construction mode (built) → Central zone: injection mode (ejected) → Outer edge: cutting mode (split)

**1. At the inner edge (Mercury, Venus, Earth):** the high rate of thermal loss causes rapid drying of the parent material. Mercury and Venus closed the geological birth window before the spin conditions were reached. Earth is located at the point where the length of the birth window allowed the unique, prolonged construction of the Moon on the tip of a pear, which dictates its high mass (1.2%) and its dry content.

**2. In the central zone (Mars to Neptune):** the bodies are massive and rich in water, and the hydrodynamic pump ejected relatively tiny moons (0.01%–0.02% in the giants), many of them water-rich, at successive stages of differentiation. The thickness of the atmospheric cap (dictated by the availability of gas in the protoplanetary disc) set the rate of water escape; on Jupiter and Saturn the thick cap trapped the water and left a corotation line that hardly moves, while on Uranus and Neptune the thin cap allowed free water escape that slowed the planets and moved the line outward.

**3. At the outer edge (Pluto):** the body is small and cold, and spun fast at that time. Internal differentiation accelerated it beyond the threshold, and brought about the split of the two-lobed structure (peanut mechanism) at its neck; Haumea has not yet split, with the dense core concentrated in the large lobe (Pluto, Triton).

The final validation column of the common-denominator matrix indicates full agreement of the geophysical picture with astronomical reality. The places where quantitative completion or exact empirical measurement is required are now isolated and gathered in the chapter of open questions (Part M).

### L.3 Who is examined: the rule for a captured moon

To prevent a methodological bias in which every celestial body that does not agree with the model's predictions would be arbitrarily classified as captured, a strict quantitative classification protocol was set in advance. A moon formed by the uniform mechanism on the corotation line must move in the direction of the parent's axial spin, in the plane of its

geometric equator, on an orbit with eccentricity close to zero (almost circular), and be made of the parent's source material.

Of these four criteria, three (eccentricity, distance and composition) may change secondarily even for an original moon owing to complex dynamical interactions (orbital stretching by a neighbour, radial push by a ring, or ejection at different stages of differentiation). By contrast, the fourth sign – a reverse orbit (Retrograde) – cannot develop for an original moon in any mechanical configuration, since a parent does not eject mass against the vector of its own spin; it is therefore given a double quantitative weight.

**The defined scoring protocol:**

- Reverse orbit (orbital inclination $> 90^{\circ}$ from the effective Laplace plane) = 2 points.
- High orbital inclination in the normal direction of rotation = 1 point.
- High eccentricity (ellipticity) = 1 point.
- Extreme radial distance that cannot be reached from the birth range by a known mechanical push = 1 point.
- Different chemical composition of the whole body (excluding secondary accretion of a surface coating) = 1 point.

Cut-off criterion: a body accumulating 3 points or more is classified as captured.

Methodological note: the orbital inclination is measured relative to the reference surface of the Laplace plane (Laplace plane) and not to the plane of the planetary equator, since on distant orbits the gravitational field of the Sun shifts the stability of the orbit to a secondary reference plane. Earth's Moon, for example, is inclined $18^{\circ} - 29^{\circ}$ from the plane of Earth's equator, but shows an inclination of only $5^{\circ}$ from the Laplace plane.

**Table applying the quantitative classification protocol to moons**

| Celestial body | Direction of orbit (2) | Eccentricity (1) | Orbital distance (1) | Chemical composition (1) | Total points | Geodynamic conclusion |
|---|---|---|---|---|---|---|
| Our Moon | 0 | 0 | 0 | 0 | 0 | Original (born) |
| Iapetus | 0 | 0 | 0 (underwent a push) | 0 (dark side = dust) | 0 | Original (born) |

| | | | | | | |
|---|---|---|---|---|---|---|
| Triton | 2 | 0 | 0 | 1 | 3 | Captured from the Kuiper belt |
| Phoebe | 2 | 0 | 1 | 1 | 4 | Captured |
| Sycorax | 2 | 1 | 1 | 1 | 5 | Captured |
| Nereid | 0 | 1 | 1 | 0 | 2 | Not captured; marked for examination |
| Himalia | 0 | 0 | 1 | 1 | 2 | Not captured; marked for examination |

The reverse-orbit signature is the cleanest empirical indicator. Gravitationally captured bodies enter the system from any potential direction; however, at large distances from the parent, the Sun's perturbing torque removes and breaks up moons moving in the normal direction of rotation, while moons on reverse orbits keep dynamical stability up to about 70% of the Hill radius (compared with only 50% for a normal orbit) [Hamilton & Burns 1991]. This mechanical configuration explains why most of the outer moons of the giants are retrograde. Bodies receiving 2 points in the matrix (such as Nereid and Himalia) are not declared captured, but remain under the system protocol checks of the host planet.

### L.4 The arc of births

The formation of a moon depends on the simultaneous existence of four physical conditions in the parent: active differentiation, a time window, an open (not sealed) crust, and a subsurface layer of low viscosity. The radial distance from the Sun dictates the length of time during which these four conditions hold together, and produces a distribution function with the morphology of an arc.

• **The inner thermal zone (Mercury, Venus):** the high rate of thermal loss causes rapid drying of the outer envelope, an extreme rise in viscosity and early sealing of the crust, before the required axial spin thresholds are reached. Mercury and Venus are moonless as a direct result. In the modern era, the corotation line of both bodies extends beyond the active boundary of their Hill radius (Venus: $1.53 \times 10^6$ km versus a Hill radius of $1.01 \times 10^6$ km; Mercury: $242,900$ km versus a Hill radius of $220,600$ km owing to a spin period of 58.6 days), a configuration that entirely prevents orbital stability. Earth almost did not give birth: it was far enough from the Sun to allow the prolonged construction stage of the Moon on the

tip of the pear, without which the material would not have reached independent orbital velocity and would have remained locked to the parent.

• **The zone of peak pumping (Mars to Saturn):** as the accretion distance from the Sun increases, the celestial bodies keep a low mean density and a high water content, which allow continuous ejection of mass. The rate of ejection of regular moons and their relative mass increase gradually, until the structural maximum is reached in the Saturn system.

• **The outer icy zone (beyond Saturn):** the low temperatures cause rapid freezing of the outer layers. A rigid, frozen crust mechanically blocks the geological ejection openings and reduces the efficiency of the hydrodynamic pump. Uranus showed a decline in the rate of ejection, and Neptune ejected only a tiny system. Pluto underwent a full hydrodynamic split in a primordial period when the core was hot and soft; in the present era the body is completely frozen and static.

To keep the analysis clean, the table and the vector model (Figure L.3) count regular moons only, entirely omitting captured bodies according to the criteria of L.3 (including Triton and Nereid). In the overall global count (including captured moons), the peak is kept in the Saturn system (Jupiter: about 97, Saturn: 274 or more, Uranus: 29, Neptune: 16), but most of the inventory in these catalogues represents secondary capture and is unrelated to the planetary birth mechanism.

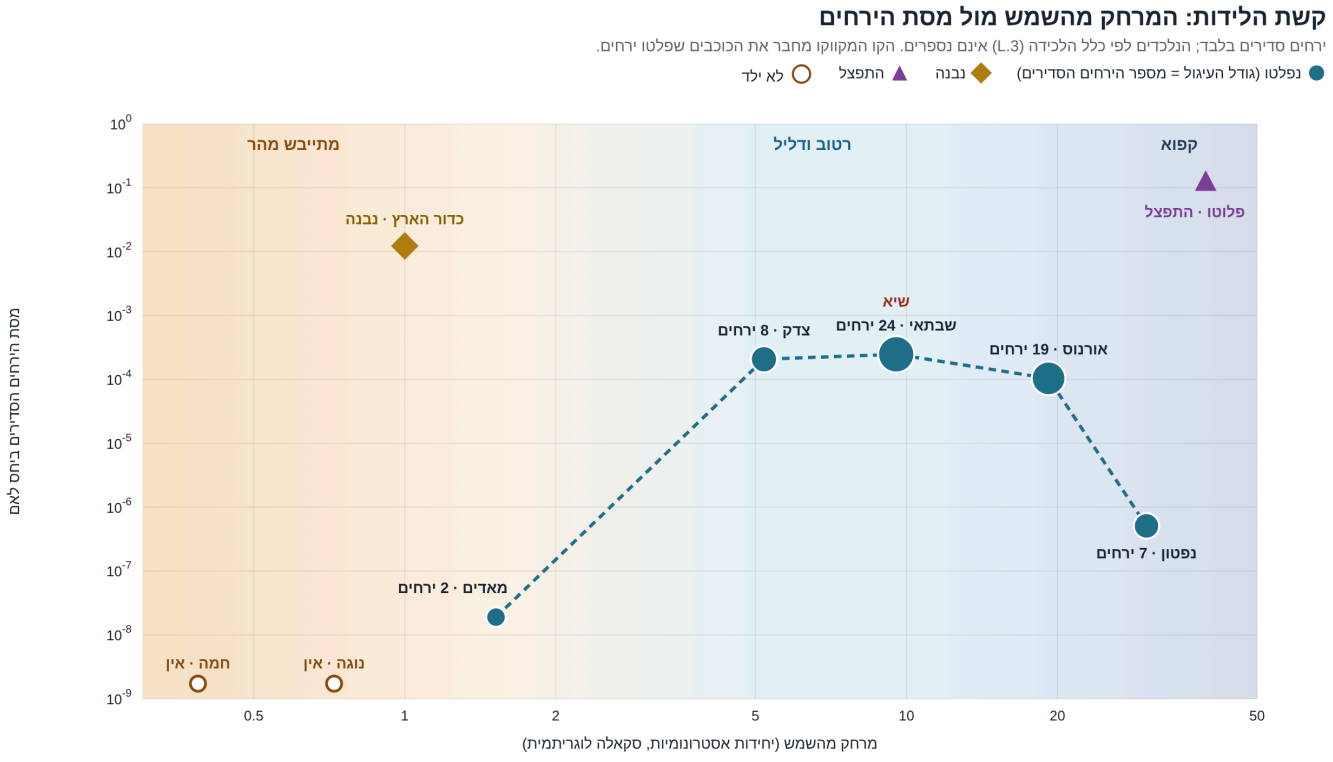



**Figure L.3.** The arc of births: distance from the Sun (horizontal axis, logarithmic) against the mass of the regular moons relative to the parent (vertical axis,

logarithmic). The dashed line connects the planets that ejected moons, and the size of the circle is the number of regular moons. Earth (built) and Pluto (split) are marked separately, because their mode of birth is different. *How to read:* for the planets that ejected moons, the mass rises up to Saturn and falls after it; from Uranus to Neptune it falls by a factor of about 200. Masses: Jacobson 2014 (Uranus), Jacobson 2010 (Mars); the masses of Neptune's inner moons have not been measured, and are estimated from their size and an assumed density.

### Parameters of the system's arc of births (regular moons only)

| Parent | Distance from the Sun ($\mathrm{AU}$) | Thermodynamic state at the birth stage | Mechanical mode of birth | Inventory of regular moons | Total mass fraction of the moons ($m_{\mathrm{moons}}/M_{\mathrm{parent}}$) |
|---|---|---|---|---|---|
| Mercury | 0.39 | Rapid thermal drying | — | 0 | 0 |
| Venus | 0.72 | Rapid thermal drying | — | 0 | 0 |
| Earth | 1.00 | Inner thermal barrier; prolonged rate of construction | Built | 1 | $1.2 \times 10^{-2}$ |
| Mars | 1.52 | Relatively dry; local bulge activity | Ejected | 2 | $1.9 \times 10^{-8}$ |
| Jupiter | 5.20 | Water-saturated, low mean density | Ejected | 8 | $2.1 \times 10^{-4}$ |
| Saturn | 9.50 | Optimal range (peak of the pump) | Ejected | about 24 | $2.5 \times 10^{-4}$ |
| Uranus | 19.20 | Onset of morphological freezing of the crust | Ejected | 19 | $1.0 \times 10^{-4}$ |
| Neptune | 30.10 | Rigid frozen crust; blocking of openings | Ejected | 7 | about $5 \times 10^{-7}$ |
| Pluto | 39.50 | Soft primordial core beneath a frozen envelope | Split | 5 | $0.12$ |

Numerical note: the moon masses are based on the data of Jacobson 2014 (Uranus) and Jacobson 2010 (Mars). Since there are no direct mass measurements for Neptune's system of inner moons, their values are calculated by integrating the body's volume with an assumed density.

Geophysical conclusion: the arc of births validates the dependence of the consolidation mechanism on radial distance from the Sun. Close to the Sun, the process is blocked thermodynamically by drying; at extreme distances, the process is weakened mechanically by the freezing of the crust and the blocking of the pump. The Saturn system is the optimal graph, showing the peak values both in the number of bodies and in their total mass fraction relative to the parent for the ejection mechanism.

## PART M Exceptions and open questions

The topics detailed below form the space of open problems that the paper leaves for further calculation and research. Each item represents a mechanical or geodynamic question arising from the application of the uniform mechanism, and all these questions, together with the lists of Part N, are gathered in Paper 13 (research topics).

### Analysis of the open questions and structural problems

| # | Geophysical topic | Definition of the problem and the open scientific question |
|---|---|---|
| 1 | Density-order anomalies on Uranus | The density of the five large moons increases as their distance from Uranus grows, the reverse of the trend in the Jupiter system. The model of differentiation stages (A.9) proposes that Titania and Oberon were ejected at stage 1 (early) and Miranda at stage 2 (peak of the pump). The open question: does this mechanism provide a full cross-section and an exhaustive explanation of the chemical variation in the system? |
| 2 | Origin of Uranus's axial tilt | The water-escape mechanism has been rejected as a cause of vector tilting (I.1). Dynamical changes in the parent's mass distribution – the residence of water and moons on distant orbits in the plane of the equator, or the inward tidal collapse of a massive primordial moon – are the mechanical direction consistent with the model, and require a vector calculation of the resonance windows. |
| 3 | Signatures and morphology of the fracture in the Pluto–Charon system | On Charon, the Vulcan Planitia plain represents the inner material exposed at the moment the neck tore (K.1). Since Pluto continued its axial spin after the split, the vector of its mass asymmetry wandered before mutual locking was reached: a mass deficit wanders toward the spin pole, and a mass excess is moved to Charon's axis of attraction (near or far side). The question: is Pluto's fracture scar located close to the geometric pole, or is the Sputnik Planitia basin (which sits almost opposite the point facing Charon) the fracture site that was filled in? In addition, the structural origin of the secondary smooth plain of about 400 km diameter in Charon's far hemisphere remains open [Beyer et al. 2021]. |

| | | |
|---|---|---|
| 4 | Morphology of the Moon's frozen shape | Quantitative verification is required as to whether the landing of returning mass (secondary accretion) explains the thickening of the crust on the far side, and whether the tensile stresses on the tip of the pear are responsible for the two-sided egg-shaped oblateness component. The decisive test: separation and integration of the two morphological components on the basis of data from the GRAIL gravimetric mission and the LOLA topographic data. |
| 5 | Quantitative reconstruction of the mass of water lost by the giants | Applying protocol check 3 in reverse: calculating the exact mass of water that had to cross the corotation line to supply the braking torque that moved the line to its modern position. A preliminary analysis sets a lower bound of 2%–3% of the mass of Uranus and Neptune. |
| 6 | Density anomalies in the Saturn system | The densities of the medium-sized moons show random fluctuations with no monotonic dependence on radial distance. The question: did the thickness of Saturn's atmospheric cap force intermittent pump activity (pulsed supply of the ejection opening), thereby causing intermittent supply and an alternation between ejections of dry material and water-saturated material? |
| 7 | Mechanical blocking of birth on Ceres (Ceres) | Ceres is rich in water components, and its corotation line (2.54R) stands close to the birth range, but the body is moonless. The question is open: does the blocking of birth arise from the length of the birth window, from the limited dimensions of the body, or from an external gravitational perturbing torque by Jupiter? |
| 8 | Dynamics of active birth on Haumea (Haumea) | The body represents a peanut structure that has not yet split, and the tips of its long axis extend beyond the corotation line. The question: is the body carrying out spontaneous mass ejection at present, and are the ring system, the moons and the dynamical family material scooped from the tips of the axis rather than the product of a giant impact? Identification of morphological signs of narrowing (a neck) is required. |
| 9 | Criteria for distinguishing between a dwarf planet and an escaped moon | Definition of absolute geodynamic boundaries between a celestial body that is an original dwarf planet and a body that is a primordial moon detached from a host system. The Pluto–Charon system proves that a dwarf pair can develop as the product of one united body that split. |
| 10 | Mechanics of the formation of equatorial ridges | The massive topographic ridge along the equator of Iapetus, as well as the geometric structures of Pan and Atlas: are these structures inner envelope material that was stretched and hardened at the moment of mechanical separation, as in the spatula-ridge model [Paper 9]? |
| 11 | Chronology of the ejection of Pluto's small moons | The bodies were released from the equator of the peanut before the final split stage (K.1). The dynamical question: was part of the inventory ejected as a |

| | | |
|---|---|---|
| | | product of the mechanical fracture of the crust at the moment the neck was cut through? The present empirical data do not allow a vector decision. |

## PART N Research list for each body

For each celestial body a specific protocol of measurements and analyses is defined, intended to test empirically the validity of the uniform mechanism and of the reconstructed parameters.

**Catalogue of research targets and geophysical measurements**

- Mercury:
  - Complete empirical exclusion of morphological signs of a bulge or pear on the surface, as final verification that the birth window did not open.
  - Vector reconstruction of the initial axial spin before the stage of tidal locking to the Sun.
- Venus:
  - Deep topographic analysis of the Aphrodite Terra highland to exclude geodynamic remnants of a primordial bulge.
  - Modelling of the history of axial spin and the evolution of the dense atmosphere.
- Earth:
  - Mathematical isolation of the asymmetric (one-sided) crustal-thickening component from the symmetric (two-sided) egg-shaped oblateness component in the data on the Moon's morphology.
  - Reconstruction of the history of the mean tidal dissipation factor ($k_2/Q$) over the geological eras.
- Mars:
  - Quantitative calculation of the angular momentum and mass removed from the system by the water components and the primordial ring.
  - Measurement of the internal structure of Phobos and Deimos to determine the exact porosity (Macro-porosity), as verification of ejection at stage 2 (peak of the pump).
  - Modelling of the survival of Phobos under a time-varying dissipation factor.
- Jupiter:
  - Application of the full protocol to the moon Himalia and its dynamical group (which holds 2 points under the capture rule L.3).

    - Cross-check between the density of the inner body (without the gas envelope) and the density of the moon system.
    - Determination of the degree of internal differentiation of each Galilean moon, as an indicator of the stage of the parent's pump activity at the moment of its birth.
    - Modelling of the dynamical mechanism that supplied the outward push for Europa, Ganymede and Callisto.
    - Validation of the chronology of the order of birth by crater dating and determination of the surface ages of the moons.
- Saturn:
    - Simultaneous measurement of the rate of water leakage from the pump to the rings and the rate of mass removal by ring rain, to validate the continuous-flow model.
    - Identification and mapping of frozen shapes (Fossil bulges) in the medium-sized moons, to reconstruct the radius of their original birth site.
    - Quantitative solution of the density-order anomalies in the medium-sized moons.
- Uranus:
    - Reverse angular-momentum balance (check 3) to determine the total mass of water that escaped through the corotation line.
    - Gravimetric analysis comparing the degree of differentiation of Titania and Oberon with that of spongy Miranda.
    - Mathematical calculation crossing the mass-distribution moment of the births and of water escape with the stages of precession and the 98° axial tilt.
- Neptune:
    - Detailed dynamical analysis of the moon Nereid (which holds 2 points under the capture rule): determining whether its extreme eccentricity is mechanically consistent with original birth from Neptune and post-birth disruption.
    - Calculation of the mass of water that escaped from the line owing to the thin atmospheric cap.
    - **Direct empirical measurement of the density of the moon Proteus (model prediction:** exceptionally low density owing to structural sponginess).
    - Reconstruction of the architecture of the original regular moon system before the capture of Triton.
- Pluto:

    - Gravimetric and topographic mapping of Pluto to locate the geodynamic fracture scar (at the spin pole or on the coupling axis to Charon).
    - High-resolution imaging of Pluto's hemisphere facing Charon (a region imaged at low resolution during the New Horizons mission).
    - Tectonic analysis of the smooth plain in Charon's far hemisphere.
    - **Measurement of Charon's moment of inertia (model prediction:** only partial internal differentiation).
- Haumea (Haumea):
    - Observational monitoring to detect ejection of material or dust from the tips of the long geometric axis.
    - Modelling of the dynamical coupling between the ring system, the moons and the orbital family.
    - Identification of morphological signatures of neck narrowing along the principal axis.
- Eris and Makemake (Eris & Makemake):
    - Exact empirical measurement of the density of the moons (Dysnomia and the secondary moon of Makemake).
    - Reconstruction of the history of the systems' rotational evolution.
- Ceres (Ceres):
    - Isolation of the mechanical parameter that blocked birth (deviation in window length, small mass dimensions, or a perturbing torque from Jupiter).

## CODE Data and code availability

All the physical, geodynamic and orbital calculations presented throughout the paper – including the positions of the dynamical corotation lines, the birth-day values, the angular-momentum reconstructions, the tidal-migration integrations and the Hill radii – were carried out with a concise Python script based on basic formulas, on the basis of the empirical data presented in the scientific sources. The calculations can be fully reproduced mathematically , and the full script together with the data matrices will be included in a structured way in the main code package of the paper series:

- **Repository:** Zenodo
- **Digital identifier (DOI):** doi:10.5281/zenodo.22743615

## REFERENCES References


Agnor, C. B., & Hamilton, D. P. (2006). Neptune's capture of its moon Triton in a binary–planet gravitational encounter. Nature, 441, 192–194.

Alemi, A., & Stevenson, D. (2006). Why Venus has no moon. Bulletin of the American Astronomical Society, 38, 491.

Anderson, J. D., Jacobson, R. A., McElrath, T. P., et al. (2001). Shape, mean radius, gravity field, and interior structure of Callisto. Icarus, 153, 157–161.

Anderson, J. D., Lau, E. L., Sjogren, W. L., Schubert, G., & Moore, W. B. (1996). Gravitational constraints on the internal structure of Ganymede. Nature, 384, 541–543.

Anderson, J. D., Schubert, G., Jacobson, R. A., et al. (1998). Europa's differentiated internal structure: inferences from four Galileo encounters. Science, 281, 2019–2022.

Anderson, J. D., Sjogren, W. L., & Schubert, G. (1996). Galileo gravity results and the internal structure of Io. Science, 272, 709–712.

Andert, T. P., Rosenblatt, P., Pätzold, M., et al. (2010). Precise mass determination and the nature of Phobos. Geophysical Research Letters, 37, L09202.

Beyer, R. A., Nimmo, F., McKinnon, W. B., et al. (2017). Charon tectonics. Icarus, 287, 161–174.

Beyer, R. A., Robbins, S. J., Beddingfield, C., et al. (2021). Charon's far side geomorphology. The Planetary Science Journal, 2, 141.

Bills, B. G., & Ray, R. D. (1999). Lunar orbital evolution: a synthesis of recent results. Geophysical Research Letters, 26, 3045–3048.

Black, B. A., & Mittal, T. (2015). The demise of Phobos and development of a Martian ring system. Nature Geoscience, 8, 913–917.

Boué, G., & Laskar, J. (2010). A collisionless scenario for Uranus tilting. The Astrophysical Journal Letters, 712, L44–L47.

Brown, M. E., & Butler, B. J. (2023). Masses and densities of dwarf planet satellites measured with ALMA. The Planetary Science Journal, 4, 193.

Brown, M. E., Barkume, K. M., Ragozzine, D., & Schaller, E. L. (2007). A collisional family of icy objects in the Kuiper belt. Nature, 446, 294–296.

Brozović, M., Showalter, M. R., Jacobson, R. A., & Buie, M. W. (2015). The orbits and masses of satellites of Pluto. Icarus, 246, 317–329.

Burns, J. A. (1973). Where are the satellites of the inner planets? Nature Physical Science, 242, 23–25.

Castillo-Rogez, J. C., Matson, D. L., Sotin, C., et al. (2007). Iapetus' geophysics: rotation rate, shape, and equatorial ridge. Icarus, 190, 179–202.

Chandrasekhar, S. (1969). Ellipsoidal Figures of Equilibrium. Yale University Press.

Cheng, W. H., Lee, M. H., & Peale, S. J. (2014). Complete tidal evolution of Pluto–Charon. Icarus, 233, 242–258.

Correia, A. C. M., & Laskar, J. (2001). The four final rotation states of Venus. Nature, 411, 767–770.

Cruikshank, D. P., Dalton, J. B., Dalle Ore, C. M., et al. (2007). Surface composition of Hyperion. Nature, 448, 54–56.

Cuzzi, J. N., Burns, J. A., Charnoz, S., et al. (2010). An evolving view of Saturn's dynamic rings. Science, 327, 1470–1475.

Dauphas, N., & Pourmand, A. (2011). Hf–W–Th evidence for rapid growth of Mars and its status as a planetary embryo. Nature, 473, 489–492.

Donahue, T. M., Hoffman, J. H., Hodges, R. R., & Watson, A. J. (1982). Venus was wet: a measurement of the ratio of deuterium to hydrogen. Science, 216, 630–633.

Garrick-Bethell, I., Wisdom, J., & Zuber, M. T. (2006). Evidence for a past high-eccentricity lunar orbit. Science, 313, 652–655.

Goldreich, P., Murray, N., Longaretti, P. Y., & Banfield, D. (1989). Neptune's story. Science, 245, 500–504.

Haisch, K. E., Lada, E. A., & Lada, C. J. (2001). Disk frequencies and lifetimes in young clusters. The Astrophysical Journal, 553, L153–L156.

Hamilton, D. P., & Burns, J. A. (1991). Orbital stability zones about asteroids. Icarus, 92, 118–131.

Iess, L., Militzer, B., Kaspi, Y., et al. (2019). Measurement and implications of Saturn's gravity field and ring mass. Science, 364, eaat2965.

Iess, L., Rappaport, N. J., Jacobson, R. A., et al. (2010). Gravity field, shape, and moment of inertia of Titan. Science, 327, 1367–1369.

Jacobson, R. A. (2010). The orbits and masses of the Martian satellites and the libration of Phobos. The Astronomical Journal, 139, 668–679.

Jacobson, R. A. (2014). The orbits of the Uranian satellites and rings, the gravity field of the Uranian system, and the orientation of the pole of Uranus. The Astronomical Journal, 148, 76.

JPL Solar System Dynamics (2024). Planetary satellite physical and mean orbital parameters. https://ssd.jpl.nasa.gov

Kivelson, M. G., Khurana, K. K., Russell, C. T., et al. (1996). Discovery of Ganymede's magnetic field by the Galileo spacecraft. Nature, 384, 537–541.

Kruijer, T. S., Burkhardt, C., Budde, G., & Kleine, T. (2017). Age of Jupiter inferred from the distinct genetics and formation times of meteorites. PNAS, 114, 6712–6716.

Lainey, V., Arlot, J.-E., Karatekin, Ö., & Van Hoolst, T. (2009). Strong tidal dissipation in Io and Jupiter from astrometric observations. Nature, 459, 957–959.

Lainey, V., Casajus, L. G., Fuller, J., et al. (2020). Resonance locking in giant planets indicated by the rapid orbital expansion of Titan. Nature Astronomy, 4, 1053–1058.

Lainey, V., Jacobson, R. A., Tajeddine, R., et al. (2017). New constraints on Saturn's interior from Cassini astrometric data. Icarus, 281, 286–296.

Lawrence, D. J., Feldman, W. C., Goldsten, J. O., et al. (2013). Evidence for water ice near Mercury's north pole from MESSENGER Neutron Spectrometer measurements. Science, 339, 292–296.

Leconte, J., & Chabrier, G. (2012). A new vision of giant planet interiors: impact of double diffusive convection. Astronomy & Astrophysics, 540, A20.

Li, C., Ingersoll, A., Bolton, S., et al. (2020). The water abundance in Jupiter's equatorial zone. Nature Astronomy, 4, 609–616.

Mankovich, C. R., & Fortney, J. J. (2020). Evidence for a dichotomy in the interior structures of Jupiter and Saturn from helium phase separation. The Astrophysical Journal, 889, 51.

Mankovich, C. R., & Fuller, J. (2021). A diffuse core in Saturn revealed by ring seismology. Nature Astronomy, 5, 1103–1109.

Mankovich, C., Marley, M. S., Fortney, J. J., & Movshovitz, N. (2019). Cassini ring seismology as a probe of Saturn's interior. I. Rigid rotation. The Astrophysical Journal, 871, 1.

Margot, J.-L., Campbell, D. B., Giorgini, J. D., et al. (2021). Spin state and moment of inertia of Venus. Nature Astronomy, 5, 676–683.

Morbidelli, A., Tsiganis, K., Batygin, K., Crida, A., & Gomes, R. (2012). Explaining why the uranian satellites have equatorial prograde orbits despite the large planetary obliquity. Icarus, 219, 737–740.

Murray, J. B., & Heggie, D. C. (2014). Character and origin of Phobos' grooves. Planetary and Space Science, 102, 119–143.

O'Donoghue, J., Moore, L., Connerney, J., et al. (2019). Observations of the chemical and thermal response of 'ring rain' on Saturn's ionosphere. Icarus, 322, 251–260.

Ortiz, J. L., Santos-Sanz, P., Sicardy, B., et al. (2017). The size, shape, density and ring of the dwarf planet Haumea from a stellar occultation. Nature, 550, 219–223.

Owen, T., Danielson, G. E., Cook, A. F., et al. (1979). Jupiter's rings. Nature, 281, 442–446.

Pappalardo, R. T., Reynolds, S. J., & Greeley, R. (1997). Extensional tilt blocks on Miranda: evidence for an upwelling origin of Arden Corona. Journal of Geophysical Research, 102, 13369–13379.

Parker, A. H., Buie, M. W., Grundy, W. M., & Noll, K. S. (2016). Discovery of a Makemakean moon. The Astrophysical Journal Letters, 825, L9.

Park, R. S., Konopliv, A. S., Bills, B. G., et al. (2016). A partially differentiated interior for (1) Ceres deduced from its gravity field and shape. Nature, 537, 515–517.

Pearl, J. C., Conrath, B. J., Hanel, R. A., et al. (1990). The albedo, effective temperature, and energy balance of Uranus, as determined from Voyager IRIS data. Icarus, 84, 12–28.

Peplowski, P. N., Evans, L. G., Hauck, S. A., et al. (2011). Radioactive elements on Mercury's surface from MESSENGER: implications for the planet's formation and evolution. Science, 333, 1850–1852.

Pollack, J. B., Hubickyj, O., Bodenheimer, P., et al. (1996). Formation of the giant planets by concurrent accretion of solids and gas. Icarus, 124, 62–85.

Porco, C. C., Thomas, P. C., Weiss, J. W., & Richardson, D. C. (2007). Saturn's small inner satellites: clues to their origins. Science, 318, 1602–1607.

Postberg, F., Kempf, S., Schmidt, J., et al. (2009). Sodium salts in E-ring ice grains from an ocean below the surface of Enceladus. Nature, 459, 1098–1101.

Rabinowitz, D. L., Barkume, K., Brown, M. E., et al. (2006). Photometric observations constraining the size, shape, and albedo of 2003 EL61, a rapidly rotating, Pluto-sized object in the Kuiper belt. The Astrophysical Journal, 639, 1238–1251.

Rogoszinski, Z., & Hamilton, D. P. (2020). Tilting ice giants with a spin–orbit resonance. The Astrophysical Journal, 888, 60.

Saillenfest, M., Rogoszinski, Z., Lari, G., Baillié, K., Boué, G., Crida, A., & Lainey, V. (2022). Tilting Uranus via the migration of an ancient satellite. Astronomy & Astrophysics, 668, A108.

Showalter, M. R., & Hamilton, D. P. (2015). Resonant interactions and chaotic rotation of Pluto's small moons. Nature, 522, 45–49.

Showalter, M. R., de Pater, I., Lissauer, J. J., & French, R. S. (2019). The seventh inner moon of Neptune. Nature, 566, 350–353.

Smith, B. A., Soderblom, L. A., Banfield, D., et al. (1989). Voyager 2 at Neptune: imaging science results. Science, 246, 1422–1449.

Stern, S. A., Bagenal, F., Ennico, K., et al. (2015). The Pluto system: initial results from its exploration by New Horizons. Science, 350, aad1815.

Stern, S. A., Weaver, H. A., Spencer, J. R., et al. (2019). Initial results from the New Horizons exploration of 2014 MU69, a small Kuiper Belt object. Science, 364, eaaw9771.

Szakáts, R., Kiss, Cs., Ortiz, J. L., et al. (2023). Tidally locked rotation of the dwarf planet (136199) Eris discovered via long-term ground-based and space photometry. Astronomy & Astrophysics, 669, L3.

Thomas, P. C., Armstrong, J. W., Asmar, S. W., et al. (2007). Hyperion's sponge-like appearance. Nature, 448, 50–53.

Verbiscer, A. J., Skrutskie, M. F., & Hamilton, D. P. (2009). Saturn's largest ring. Nature, 461, 1098–1100.

Verbiscer, A., French, R., Showalter, M., & Helfenstein, P. (2007). Enceladus: cosmic graffiti artist caught in the act. Science, 315, 815.

Wahl, S. M., Hubbard, W. B., Militzer, B., et al. (2017). Comparing Jupiter interior structure models to Juno gravity measurements and the role of a dilute core. Geophysical Research Letters, 44, 4649–4659.

Waite, J. H., Perryman, R. S., Perry, M. E., et al. (2018). Chemical interactions between Saturn's atmosphere and its rings. Science, 362, eaat2382.

Ward, W. R., & Hamilton, D. P. (2004). Tilting Saturn. I. Analytic model. The Astronomical Journal, 128, 2501–2509.

Ward, W. R., & Canup, R. M. (2006). Forced resonant migration of Pluto's outer satellites by Charon. Science, 313, 1107–1109.

Ward, W. R., & Reid, M. J. (1973). Solar tidal friction and satellite loss. Monthly Notices of the Royal Astronomical Society, 164, 21–32.

Weaver, H. A., Buie, M. W., Buratti, B. J., et al. (2016). The small satellites of Pluto as observed by New Horizons. Science, 351, aae0030.

Williams, D. R. (2024). Planetary fact sheets. NASA Goddard Space Flight Center. https://nssdc.gsfc.nasa.gov/planetary/factsheet/

Wilson, H. F., & Militzer, B. (2012). Rocky core solubility in Jupiter and giant exoplanets. Physical Review Letters, 108, 111101.

Wisdom, J., Dbouk, R., Militzer, B., et al. (2022). Loss of a satellite could explain Saturn's obliquity and young rings. Science, 377, 1285–1289.

Wisdom, J., Peale, S. J., & Mignard, F. (1984). The chaotic rotation of Hyperion. Icarus, 58, 137–152.

Young, E. D., Kohl, I. E., Warren, P. H., et al. (2016). Oxygen isotopic evidence for vigorous mixing during the Moon-forming giant impact. Science, 351, 493–496.

Zahnle, K., Schenk, P., Levison, H., & Dones, L. (2003). Cratering rates in the outer Solar System. Icarus, 163, 263–289.

Duran, Z. (2026). The LEF Model series, Papers 1–13. Zenodo. doi:10.5281/zenodo.22743615 and lef-model.com.

## DECLARATIONS Declarations

Use of artificial intelligence. The author used Claude (Anthropic) for calculations, for gathering data from the sources, for building the figures and for wording. The ideas and statements of the model are the author's; insights that arose from the calculations are marked as such in the paper's work log. All data were checked against the cited sources.

Funding and conflict of interest. The research was carried out without external funding. The author has no conflict of interest.

---

© 2026 Zohar Duran. CC BY 4.0. lef-model.com